

The Kronecker character χ_{-4} as the joint source of the lemniscatic identity algebra:

a unified treatment of $G^* = \Gamma(1/4)/\Gamma(3/4)$ and $G_G = 1/\text{AGM}(1, \sqrt{2})$

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Abstract

The lemniscatic CM point $\tau = i$ generates a single transcendental constant whose value has two equally-natural normalisations: the *ratio of the Euler reflection at $z = 1/4$* , $G^* := \Gamma(1/4)/\Gamma(3/4) \approx 2.95867\dots$, and *Gauss's constant*, $G_G := 1/\text{AGM}(1, \sqrt{2}) \approx 0.83463\dots$. They are linearly related by $G^* = 2\sqrt{\pi} G_G$ (see Figure 1), and either may be chosen as the principal variable. We show that the choice is not arbitrary: most identities in this neighbourhood have a preferred *natural form* with integer or rational coefficients in one of $\{G^*, G_G\}$, while a small number of boundary-period identities (notably the real period $\omega_E = G^* \sqrt{\pi} = 2\pi G_G$ and the Beta integral $B(1/4, 1/4) = \sqrt{2\pi} G^* = 2\sqrt{2} \pi G_G$) are clean in both. Specifically, identities arising from the *algebraic* structure (Euler reflection, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\text{Aut}(E)|^2$ for the lemniscatic curve $E : y^2 = x^3 - x$) are natural in G^* ; identities arising from the *analytic* structure (modular forms at $\tau = i$, the body-centred-cubic Watson integral, AGM iteration, the Dedekind eta function) are natural in G_G .

We present this dichotomy as the organising principle of the paper. We catalogue over fifty identities in their natural form, including the clean evaluations

$$\begin{aligned} E_4(i) &= 3 G_G^4, & \Delta(i) &= G_G^{12}/64, & \eta(i) &= G_G^{1/2}/2^{1/4}, \\ W_{\text{BCC}}^{(3)} &= 2 G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2, & W_{\text{BCC}}^{(D)} &= {}_D F_{D-1}(\tfrac{1}{2}, \dots; 1, \dots; 1), \end{aligned}$$

the reflection-form identities $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ and $\Gamma(1/4)^4 = 2\pi^2 G^{*2}$, and the master quadratic $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$ whose integer coefficient 16 is the squared automorphism count of the lemniscatic curve.

Three structural results emerge: **(I)** the value algebra of quasi-modular forms at $\tau = i$ is the polynomial ring $\mathbb{Q}[\pi^{-1}, G_G^4]$ in two algebraically independent generators under Chudnovsky's theorem, with π^{-1} recording the quasi-modular anomaly and G_G^4 recording the modular weight; **(II)** the $m \mid k$ vanishing principle, stating that at a CM point with $|\text{Aut}(E)| = m$ a modular form f of weight k satisfies $f(\tau_{\text{CM}}) = 0$ unless $m \mid k$, holds in the only two non-trivial cases over $\mathbb{Q}$ (lemniscatic with $m = 4$, equianharmonic with $m = 6$); **(III)** the equianharmonic Gauss-analog G_ρ admits the cubic-AGM expression $G_\rho = (2/\sqrt{3})/M_3(1, 2^{-1/3})$ at the cubic self-dual point $x^3 = 1/2$, completing the structural parallel with the lemniscatic $G_G = 1/\text{AGM}(1, \sqrt{2})$ at the self-complementary modulus $k = 1/\sqrt{2}$. Every identity is traced to its earliest source; the systematic dual-constant presentation appears to be new.

A separate note in preparation by the same author draws consequences of the χ_{-4} -unification developed here for a discrete-substrate physics framework; the present paper is independent of and does not address that work.

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1 Introduction

The Gauss–Euler reflection formula

$$\Gamma(z) \Gamma(1 - z) = \frac{\pi}{\sin \pi z}, \quad z \notin \mathbb{Z}, \quad (1)$$

relates two evaluations of the Gamma function symmetric about $z = 1/2$. At $z = 1/4$ the product

$$\Gamma(1/4) \Gamma(3/4) = \pi \sqrt{2}$$

is the unique linear combination of $\Gamma(1/4)$ and $\Gamma(3/4)$ that reduces to a closed form in π . The other natural combination is the ratio,

$$G^* := \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511918863889231 \dots \quad (2)$$

which yields a new transcendental.

Independently of (1), the arithmetic–geometric mean of Gauss provides a second natural lemniscatic CM constant:

$$G_G := \frac{1}{\text{AGM}(1, \sqrt{2})} = 0.83462684167407318628 \dots \quad (3)$$

known classically as *Gauss’s constant*. A brief history clarifies its standing: Gauss recorded the central observation $\text{AGM}(1, \sqrt{2}) \cdot \varpi / \pi = 1$ (equivalent to $\text{AGM}(1, \sqrt{2}) = \pi / \varpi$) in his diary on 30 May 1799 [13, Werke III, p. 370], calling the discovery a result that “*will surely open an entirely new field of analysis*.” He kept the result essentially unpublished during his lifetime; the AGM theory appeared posthumously in his *Werke*, with most details remaining in his *Nachlass* until Geppert’s 1927 edition. Brent [2] and Salamin [3] independently rediscovered the AGM-based algorithm for π in 1976, recognising the connection to Gauss’s diary. The constant $G_G = 1/\text{AGM}(1, \sqrt{2})$ now bears Gauss’s name in standard references [4, 5]. The lemniscate constant $\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\pi) \approx 2.62206 \dots$ equals πG_G , and connects (2) and (3) by the chain of identities (proved in §2):

$$G^* = 2\sqrt{\pi} G_G, \quad \varpi = \pi G_G = \frac{G^* \sqrt{\pi}}{2}. \quad (4)$$

Either G^* or G_G may be chosen as the principal variable in this neighbourhood, but the choice has a clear structural fingerprint, and that fingerprint is the central observation of the present paper:

Observation (Dichotomy of the natural form). Identities arising from the *algebraic* structure of the lemniscatic CM point—reflection ratios, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\text{Aut}(E)|^2$, the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$ —have rational coefficients in G^* . Identities arising from the *analytic* structure—modular forms at $\tau = i$, the body-centred-cubic lattice Green’s function, AGM iteration, the Dedekind eta function—have rational coefficients in G_G . Most identities have a preferred natural form in one of the two normalisations, with the other introducing transcendental factors of π ; a small number of boundary-period identities (notably ω_E and $B(1/4, 1/4)$, both with integer-coefficient cleanness in G^* and G_G) are exceptions where both forms are natural.

This dichotomy is structurally robust: $E_4(i) = 3 G_G^4$ is integer-coefficient clean, while $E_4(i) = 3 G^{*4}/(16\pi^2)$ requires a π^2 denominator. The master quadratic $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$ has integer coefficients in G^* because 16 is the squared automorphism count of $E : y^2 = x^3 - x$ (an algebraic invariant of the lemniscatic curve); the same polynomial written in G_G is $x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3 = 0$, which obliterates the integer structure with fractional π -powers.

This paper presents a systematic survey of identities at $\tau = i$ under this dual-constant discipline. We catalogue over fifty identities, each in its natural form, organised by the algebraic–analytic dichotomy. The component identities span the literature from Gauss (1799), Euler (1771), Watson (1939), Joyce (1971–72), Chowla–Selberg (1967), Borwein–Borwein (1987), Kontsevich–Zagier (2001), and Guttman (2010); we trace each to its earliest source. The synthetic contribution is the dual-constant framing as an organising principle, which appears not to have been stated explicitly elsewhere.

Structure of the paper. §2 establishes the bridge identity (4). §3–8 treat the algebraic side: reflection identities (§3), the three equivalent definitions of G^* (§4), the lemniscatic elliptic curve (§5), the master quadratic (§6), Beta integrals (§7), and the family R_n (§8). §9–12 treat the analytic side: the Watson lattice identity (§9), theta functions at $\tau = i$ (§10), the Dedekind eta function (§11), and modular forms at $\tau = i$ (§12, including the quasi-modular extension giving structural result (I) of the abstract). §13 addresses the Kontsevich–Zagier period status; §14 records transcendence and algebraic-independence status. §15 consolidates the dichotomy in a side-by-side compendium. §16 develops the equianharmonic dichotomy at $\tau = \rho$, yielding structural result (II) of the abstract. §17 presents the unifying χ_{-4} structure underlying both dichotomies, including the four-level projection theorem, three negative tests on the polynomial-assembly question, the asymptotic-regime analysis, the symmetric-algebra conjecture, and the minimal generating structure $\chi_{-4}(n) = \text{Im}(i^n)$. §18 states one open problem (KZ-strict status of G^* and G_G , a frontier KZ-programme question) and resolves a second item from earlier drafts (cubic-AGM expression for G_ρ , Theorem 18.1). §19 states two algebraic-independence conjectures supported by high-precision PSLQ evidence ($W_{\text{BCC}}^{(4)}$, Catalan’s constant).

2 The bridge $G^* = 2\sqrt{\pi} G_G$

Theorem 2.1. $G^* = 2\sqrt{\pi} G_G$.

Proof. By the Gauss–Legendre identity [1, Thm. 1.1], $K(k) = \pi/[2 \text{AGM}(1, \sqrt{1-k^2})]$. At $k = 1/\sqrt{2}$ (the self-complementary modulus), $\sqrt{1-k^2} = 1/\sqrt{2}$, so

$$K(1/\sqrt{2}) = \frac{\pi}{2 \text{AGM}(1, 1/\sqrt{2})} = \frac{\pi}{\sqrt{2} \text{AGM}(1, \sqrt{2})} = \frac{\pi G_G}{\sqrt{2}},$$

where we used the homogeneity $\text{AGM}(\lambda a, \lambda b) = \lambda \text{AGM}(a, b)$ with $\lambda = 1/\sqrt{2}$. The classical value $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ [1, Eq. 2.5.4] gives

$$G_G = \frac{\sqrt{2} K(1/\sqrt{2})}{\pi} = \frac{\sqrt{2} \Gamma(1/4)^2}{4\pi^{3/2}}.$$

By the reflection identity (Lemma 3.1 below), $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$, so

$$G_G = \frac{\sqrt{2} \cdot \pi\sqrt{2} G^*}{4\pi^{3/2}} = \frac{G^*}{2\sqrt{\pi}}. \quad \square$$

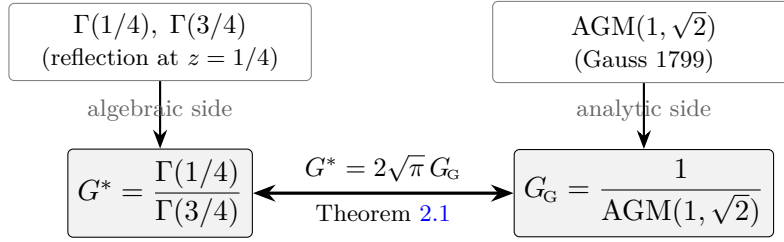


Figure 1: Two natural sources, one transcendental, two normalisations. The lemniscatic CM point generates a single value reachable from either the *algebraic* Gamma-reflection route (giving G^*) or the *analytic* AGM route (giving G_G); the two normalisations are related by the bridge $G^* = 2\sqrt{\pi} G_G$. The dichotomy thesis of this paper is that identities arising on the left side are clean in G^* , identities on the right side are clean in G_G , with a small number of borderline-period identities clean in both.

Corollary 2.2 (Conversion table). *The following identities convert between G^* , G_G , and the lemniscate constant ϖ :*

$G^* = 2\sqrt{\pi} G_G$	$G_G = G^*/(2\sqrt{\pi})$
$\varpi = \pi G_G = G^* \sqrt{\pi}/2$	$G_G = \varpi/\pi$
$G^{*2k} = (4\pi)^k G_G^{2k}$	$G_G^{2k} = G^{*2k}/(4\pi)^k$
$G^{*2k+1} = 2^{2k+1} \pi^{k+1/2} G_G^{2k+1}$	$G_G^{2k+1} = G^{*2k+1}/(2^{2k+1} \pi^{k+1/2})$

Throughout the paper, we adopt the discipline: *express each identity in the convention in which its coefficients are rational*, with the bridge identity supplied where conversion is needed. This is the operational content of Observation 1.

3 Reflection identities: the algebraic core in G^*

Lemma 3.1 (Reflection at $z = 1/4$).

$$\Gamma(1/4)^2 = \pi\sqrt{2} G^*, \quad (5)$$

$$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*, \quad (6)$$

$$\Gamma(1/4)^4 = 2\pi^2 G^{*2}, \quad (7)$$

$$\Gamma(1/4)^8 = 4\pi^4 G^{*4}. \quad (8)$$

Proof. Euler's reflection (1) at $z = 1/4$: $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$. Multiply by G^* for (5); divide for (6); square for (7) and (8). $\square$

Remark. These four identities exhibit the cleanness of G^* for the reflection-channel structure: rational coefficients in G^* (no π inside the coefficient), with π appearing only as the external factor from Euler reflection. The corresponding identities in G_G have the form $\Gamma(1/4)^2 = (\pi\sqrt{2})(2\sqrt{\pi})G_G = 2\sqrt{2}\pi^{3/2}G_G$, which introduces $\pi^{3/2}$ — fractional powers.

Corollary 3.2. *The two complementary identities are:*

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2} \quad (\text{product channel, yields } \pi), \quad (9)$$

$$\Gamma(1/4)/\Gamma(3/4) = G^* \quad (\text{ratio channel, yields the new transcendental}). \quad (10)$$

The pair $(\Gamma(1/4), \Gamma(3/4))$ is uniquely determined by these two quantities:

$$\Gamma(1/4) = \sqrt{\pi\sqrt{2}G^*}, \quad \Gamma(3/4) = \sqrt{\pi\sqrt{2}/G^*}.$$

4 Three equivalent definitions of G^*

Theorem 4.1 (Equivalent definitions). *The following are equal:*

- (i) $G^* = \Gamma(1/4)/\Gamma(3/4)$ (*Gamma-ratio form*).
- (ii) $G^* = \sqrt{2\pi}/\text{AGM}(1, 1/\sqrt{2})$ (*AGM form*).
- (iii) $G^* = \frac{1}{\sqrt{2\pi}} \int_0^1 \frac{dt}{[t(1-t)]^{3/4}}$ (*Beta-integral form*).

Proof. (i) = (iii): The Beta integral gives $B(1/4, 1/4) = \Gamma(1/4)^2/\sqrt{\pi}$. By (5), $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$, so $B(1/4, 1/4) = \sqrt{2\pi}G^*$, equivalent to (iii).

(i) = (ii): By homogeneity $\text{AGM}(\lambda a, \lambda b) = \lambda \text{AGM}(a, b)$, $\text{AGM}(1, 1/\sqrt{2}) = \text{AGM}(\sqrt{2}, 1)/\sqrt{2} = \text{AGM}(1, \sqrt{2})/\sqrt{2}$. By Theorem 2.1, $\text{AGM}(1, \sqrt{2}) = 1/G_G = 2\sqrt{\pi}/G^*$, so $\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$, which gives (ii). $\square$

Remark. Definition (i) is the algebraic-reflection form. Definition (ii) gives the bridge to G_G : rewriting (ii), $G^* = \sqrt{2\pi} \cdot \sqrt{2} \cdot G_G = 2\sqrt{\pi}G_G$, which recovers (4). Definition (iii) is the period-integral form relevant for Kontsevich–Zagier (§13). The AGM definition (ii) also admits a Borwein-like self-replicating algorithmic generalisation that computes G^* (via $\Gamma(1/4)$) to arbitrary precision; see Guillera [29] for the self-replication framework that extends the Borwein algorithms for π [1] to Γ -values at rational arguments including $\Gamma(1/4)$ and $\Gamma(1/3)$.

5 The lemniscatic elliptic curve

Definition 5.1. The lemniscatic elliptic curve over $\mathbb{Q}$ is $E : y^2 = x^3 - x$.

Lemma 5.2. *E has complex multiplication by $\mathbb{Z}[i]$, j -invariant 1728, discriminant 64, and $|\text{Aut}_{\mathbb{C}}(E)| = 4$, with the automorphism group generated by $(x, y) \mapsto (-x, iy)$.*

Theorem 5.3 (Real period). *The real period of E admits the four equivalent expressions:*

$$\omega_E = 2 \int_{-1}^0 \frac{dx}{\sqrt{x(x-1)(x+1)}} \quad (11)$$

$$= \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \quad (\text{Gamma form}) \quad (12)$$

$$= G^* \sqrt{\pi} \quad (\text{reflection form, in } G^*) \quad (13)$$

$$= 2\pi G_G \quad (\text{AGM form, in } G_G). \quad (14)$$

All four forms have rational or simple algebraic coefficients. The G^* -form and G_G -form are both clean: ω_E is a borderline identity natural in either normalisation (see Observation 5).

Proof. The integral evaluation $\omega_E = \Gamma(1/4)^2/\sqrt{2\pi}$ is classical; see [1, §2.5]. The G^* -form follows from $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ (Lemma 3.1). The G_G -form follows from the bridge identity $G^* = 2\sqrt{\pi} G_G$: $\omega_E = G^* \sqrt{\pi} = (2\sqrt{\pi} G_G) \sqrt{\pi} = 2\pi G_G$. $\square$

Observation. Theorem 5.3 reveals that $\omega_E = 2\pi G_G$ has *clean integer coefficient* in G_G (the period is literally 2π times Gauss’s constant) but *also* has clean form $G^* \sqrt{\pi}$. This is one of the borderline identities where both conventions are reasonable. The lemniscatic period sits exactly at the algebraic-analytic boundary: it is the period of an algebraic curve (algebraic origin, hence G^* representation $\omega_E = G^* \sqrt{\pi}$ is natural) but also 2π times the reciprocal of $\text{AGM}(1, \sqrt{2})$ (analytic origin, hence $\omega_E = 2\pi G_G$). Both forms are integer-coefficient.

6 The master quadratic: the algebraic centerpiece

Consider the polynomial

$$P_{G^*}(x) := x^2 - 16 G^{*2} x + 16 G^{*3} \quad (15)$$

with integer coefficient 16.

Theorem 6.1 (Master quadratic structure). *The polynomial $P_{G^*}(x)$ has positive discriminant $256 G^{*4} - 64 G^{*3} = 64 G^{*3}(4G^* - 1) > 0$ and roots*

$$x_{\pm} = 8 G^{*2} \pm 4 G^{*3/2} \sqrt{4G^* - 1}.$$

Numerically:

$$x_+ = 137.03617145815548388 \dots, \quad (16)$$

$$x_- = 3.02396391633902100 \dots \quad (17)$$

Vieta’s identities give

$$\frac{1}{x_+} + \frac{1}{x_-} = \frac{1}{G^*}, \quad x_+ \cdot x_- = 16 G^{*3}. \quad (18)$$

Proof. Direct computation via the quadratic formula and Vieta’s identities. $\square$

Observation (Integer-coefficient signature). The polynomial P_{G^*} has integer coefficients in G^* . The coefficient $16 = |\text{Aut}_{\mathbb{C}}(E)|^2$ is the squared automorphism count of the lemniscatic curve. In G_G -coordinates, the same polynomial becomes

$$x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3,$$

with explicit π and $\pi^{3/2}$ in the coefficients. The integer 16 disappears; the algebraic identification with $|\text{Aut}(E)|^2$ is obscured. This is the cleanest illustration of Observation 1: the master quadratic is intrinsically a G^* object.

Remark (Numerical content of the roots). The roots take the numerical values $x_+ \approx 137.036$ and $x_- \approx 3.024$. Their identification with physical observables and the resulting math–physics bridge lie outside the mathematical scope of the present paper and are treated in a separate note in preparation. The transcendence and algebraic-independence implications of the roots are addressed in §14.

7 Beta integrals at quarter arguments

Lemma 7.1. *The Beta function at $(1/4, 1/4)$ admits the closed forms:*

$$B(1/4, 1/4) = \int_0^1 [t(1-t)]^{-3/4} dt \quad (19)$$

$$= \frac{\Gamma(1/4)^2}{\Gamma(1/2)} = \frac{\Gamma(1/4)^2}{\sqrt{\pi}} \quad (20)$$

$$= \sqrt{2\pi} G^* \quad (\text{in } G^*, \text{ clean}) \quad (21)$$

$$= 2\sqrt{2}\pi G_G \quad (\text{in } G_G, \text{ integer-coefficient}). \quad (22)$$

Proof. The Beta-integral evaluation is classical: $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$. At $p = q = 1/4$, $\Gamma(p+q) = \Gamma(1/2) = \sqrt{\pi}$. The G^* -form follows from (5). The G_G -form follows from (4): $\sqrt{2\pi} G^* = \sqrt{2\pi} \cdot 2\sqrt{\pi} G_G = 2\sqrt{2\pi^2} G_G = 2\sqrt{2}\pi G_G$. $\square$

Remark. This is another borderline identity where both forms are clean: G^* gives $\sqrt{2\pi}$ as coefficient (irrational but classical), G_G gives 4π (integer multiple of π). Both are rational-coefficient identities in their respective constants. The Beta integral, like the lemniscatic period (5.3), sits at the boundary.

8 The family R_n and the distinguishedness of $R_4 = G^*$

Definition 8.1. For integer $n \geq 2$, $R_n := \Gamma(1/n)/\Gamma((n-1)/n)$.

Proposition 8.2. *The family $\{R_n\}_{n \geq 2}$ satisfies:*

- (i) $R_n > 0$ for all $n \geq 2$;
- (ii) $R_n = \Gamma(1/n)^2 \sin(\pi/n)/\pi$ (by Euler reflection);
- (iii) $R_2 = 1$ (trivially) and $R_n = n - 2\gamma + O(1/n)$ as $n \rightarrow \infty$, where γ is the Euler–Mascheroni constant.

The first six values:

n	R_n	associated CM-curve interpretation
2	1.0000000000	trivial
3	1.9783642596...	equianharmonic $E : y^2 = x^3 - 1$, $\text{End}(E) = \mathbb{Z}[\rho]$, $ \text{Aut} = 6$
4	2.9586751192... (G^*)	lemniscatic $E : y^2 = x^3 - x$, $\text{End}(E) = \mathbb{Z}[i]$, $ \text{Aut} = 4$
5	3.9432456137...	no CM curve with this argument structure
6	4.9312366764...	equianharmonic via twist $y^2 = x^3 + 1$
8	6.9140781897...	no CM curve

Theorem 8.3 (Algebraic distinguishedness of R_4). *Among the R_n family with $n \in \{2, 3, 4, 5, 6, 8, 12\}$, the value $R_4 = G^*$ is the unique value such that R_n is the gamma-ratio of a class-number-one CM elliptic curve over $\mathbb{Q}$ with $|\text{Aut}_{\mathbb{C}}(E)|^2 = 16$. The equianharmonic curve E_ρ at R_3 (and its twist at R_6) has $|\text{Aut}_{\mathbb{C}}(E_\rho)|^2 = 36$; no other R_n in the listed range corresponds to a CM elliptic curve with extra automorphisms.*

Proof. Direct enumeration. The class-number-one imaginary-quadratic fields with unit group strictly larger than $\{\pm 1\}$ are exactly $\mathbb{Q}(i)$ (units of order 4) and $\mathbb{Q}(\rho)$ (units of order 6). These yield the lemniscatic curve $y^2 = x^3 - x$ over $\mathbb{Q}$, with $R_4 = \Gamma(1/4)/\Gamma(3/4)$ and $|\text{Aut}|^2 = 16$, and the equianharmonic curve $y^2 = x^3 - 1$ (with twist $y^2 = x^3 + 1$), with $R_3 = \Gamma(1/3)/\Gamma(2/3)$ (resp. R_6) and $|\text{Aut}|^2 = 36$. For $n \in \{2, 5, 8, 12\}$, no CM elliptic curve over $\mathbb{Q}$ has R_n as its gamma-ratio. $\square$

Observation (Numerical content of the roots at R_4 and R_3). Numerical evaluation of $x_{\pm}(R_n) = (16R_n^2 \pm \sqrt{(16R_n^2)^2 - 64R_n^3})/2$ for the CM-relevant values R_3, R_4, R_6 yields:

n	R_n	$x_+(R_n)$	$x_-(R_n)$
3	1.97836	138.894 ...	2.007 ...
4	2.95868 (G^*)	137.036 ...	3.024 ...
6	4.93124	384.078 ...	4.995 ...

(In rows 3,6 the polynomial coefficient is $16R_n^2/16R_n^3$ as in the lemniscatic case; the corresponding equianharmonic master quadratic $y^2 - 36R_n^2y + 36R_n^3$ would have $|\text{Aut}|^2 = 36$ as its integer prefactor.) Their interpretation as physical observables is addressed in a separate note in preparation.

Proposition 8.4 (Family-wide master-quadratic root asymptotic, with full expansion). *For the master-quadratic family $x^2 - 16R^2x + 16R^3 = 0$ with $R > 1/4$, the smaller root admits the absolutely convergent expansion*

$$x_-(R) = \sum_{n=1}^{\infty} d_n R^{2-n}, \quad d_n = \frac{\binom{2n}{n}}{(2n-1) 2^{4n-3}},$$

giving explicit leading terms

$$x_-(R) = R + \frac{1}{16} + \frac{1}{128R} + \frac{5}{4096R^2} + \frac{7}{32768R^3} + O(R^{-4}).$$

The larger root satisfies $x_+(R) = 16R^2 - x_-(R)$ by Vieta, hence $x_+(R) = 16R^2 - R - 1/16 - 1/(128R) - O(R^{-2})$.

Proof. Write the quadratic formula in the form $x_-(R) = 8R^2(1 - \sqrt{1-y})$ with $y = 1/(4R) \in (0, 1)$. The Newton binomial expansion gives $1 - \sqrt{1-y} = \sum_{n \geq 1} c'_n y^n$ with $c'_n = \binom{2n}{n}/((2n-1)4^n)$ for $n \geq 1$, so

$$x_-(R) = 8R^2 \sum_{n \geq 1} \frac{c'_n}{(4R)^n} = \sum_{n \geq 1} \frac{8c'_n}{4^n} R^{2-n} = \sum_{n \geq 1} d_n R^{2-n},$$

where $d_n := 8c'_n/4^n = \binom{2n}{n}/((2n-1)2^{4n-3})$. Direct evaluation: $d_1 = 1$, $d_2 = 1/16$, $d_3 = 1/128$, $d_4 = 5/4096$, $d_5 = 7/32768$, recovering the leading R and the four subsequent correction terms. Convergence for $4R > 1$ is by ratio test: $|d_{n+1} R^{-1}/d_n| \rightarrow 1/(4R) < 1$. $\square$

Remark (Numerical verification across R_n). The difference $x_-(R_n) - R_n$ approaches $1/16 = 0.0625$ from above:

n	R_n	$x_-(R_n)$	$x_-(R_n) - R_n$
3	1.97836	2.04516	0.06680
4	2.95868 (G^*)	3.02396	0.06528
5	3.94325	4.00781	0.06456
6	4.93124	4.99537	0.06413
8	6.91408	6.97773	0.06366
12	10.89429	10.95752	0.06323
∞	$(n - 2\gamma)$	$(\rightarrow R_n + 1/16)$	$1/16 = 0.06250$

The constant excess $x_-(R_n) - R_n \rightarrow 1/16$ as $n \rightarrow \infty$ is a structural feature of the family $x^2 - 16R^2x + 16R^3 = 0$ and is independent of any physical interpretation of the roots (Figure 2).

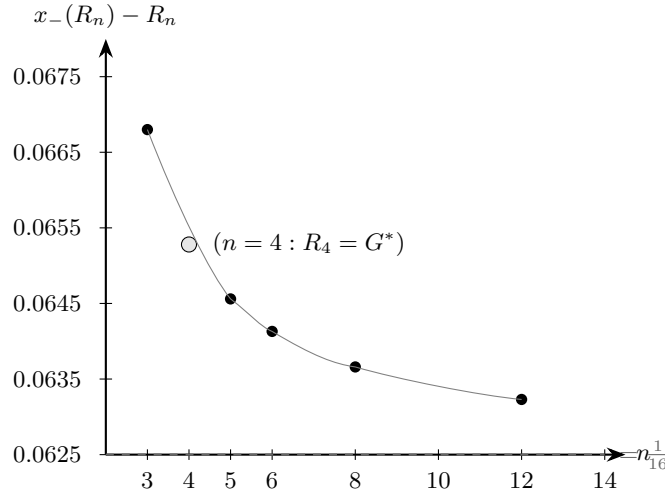


Figure 2: Numerical convergence $x_-(R_n) - R_n \rightarrow 1/16$ from above for $n \in \{3, 4, 5, 6, 8, 12\}$. The asymptote at $1/16 \approx 0.0625$ (dashed) is the structural feature of the family $x^2 - 16R^2x + 16R^3 = 0$ identified in Proposition 8.4: the leading correction $d_2 = 1/16$ is constant in R , with subsequent corrections of order $R^{-1}, R^{-2}, \dots$. The lemniscatic point $R_4 = G^*$ is highlighted; its value $x_-(G^*) - G^* \approx 0.06528$ is the $n = 4$ instance of the family-wide asymptotic.

9 The Watson lattice identity, in G_G

We now cross the algebraic–analytic boundary. The body-centred-cubic lattice Green’s function at the origin admits the cleanest expression in G_G :

Theorem 9.1 (Watson 1939, restated in G_G). *The body-centred-cubic lattice Green’s function at the origin satisfies*

$$W_{\text{BCC}}^{(3)} := \frac{1}{\pi^3} \int_0^\pi \int_0^\pi \int_0^\pi \frac{dk_1 dk_2 dk_3}{1 - \cos k_1 \cos k_2 \cos k_3} = 2 G_G^2 = \frac{2}{\text{AGM}(1, \sqrt{2})^2}. \quad (23)$$

Observation. The form $W_{\text{BCC}}^{(3)} = 2/\text{AGM}(1, \sqrt{2})^2$ states directly that the BCC random walk’s expected returns-to-origin count is computable by Gauss’s AGM applied to $(1, \sqrt{2})$, with no occurrence of π or Γ . The standard form is $\Gamma(1/4)^4/(4\pi^3)$ [18, 17]; the intermediate form $W_{\text{BCC}}^{(3)} =$

$(4/\pi^2) K(1/\sqrt{2})^2$ is also classical (cf. [1] for the AGM substitution $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$). The author has not located the fully-reduced AGM-form (23) in the surveyed literature (Watson 1939, van Peype 1938, Joyce 1971, Borwein–Borwein 1987, Guttman 2010), but does not claim it is novel; the substitution is one step from the references cited.

Proof. The classical reduction [18] gives $W_{\text{BCC}}^{(3)} = (4/\pi^2) K(1/\sqrt{2})^2$. The AGM evaluation $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$ (Theorem 11.7) then yields

$$W_{\text{BCC}}^{(3)} = \frac{4}{\pi^2} \cdot \frac{\pi^2 G_G^2}{2} = 2 G_G^2. \quad \square$$

Theorem 9.2 (Generalised Watson identity; Guttman 2010). *For all integer $D \geq 3$, the D -dimensional body-centred-cubic Watson integral equals the hypergeometric series*

$$W_{\text{BCC}}^{(D)} = {}_D F_{D-1} \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2}; 1, 1, \dots, 1; 1 \right) = \sum_{n=0}^{\infty} \binom{2n}{n}^D \frac{1}{16^n} \cdot \frac{1}{2^{n(D-2)}}. \quad (24)$$

Proof. Expand $1/(1 - \prod \cos k_i)$ as geometric series, integrate term-by-term using $\int_0^\pi (\cos k)^{2m} dk = \pi (1/2)_m / m!$, and recognise the resulting series as the hypergeometric. Convergence at $z = 1$ requires $D \geq 3$. The derivation in this form is due to Guttman [14, §2.1], who also gives the equivalent multinomial expression. $\square$

Corollary 9.3 (Numerical values, $D = 3, 4, 5$). *To 20 decimal digits:*

$$\begin{aligned} W_{\text{BCC}}^{(3)} &= 2 G_G^2 = 1.39320392968567685918 \dots, \\ W_{\text{BCC}}^{(4)} &= {}_4 F_3 \left(\frac{1}{2}; 1^3; 1 \right) = 1.11863638716418706835 \dots, \\ W_{\text{BCC}}^{(5)} &= {}_5 F_4 \left(\frac{1}{2}; 1^4; 1 \right) = 1.04682554983350000525 \dots \end{aligned}$$

Remark. The $D = 3$ case admits the closed form $2 G_G^2$; the cases $D \geq 4$ do not admit any known closed form in elementary functions or values of Γ . PSLQ search to 30+ digit precision finds no integer relation between $W_{\text{BCC}}^{(4)}$ and any rational power product of $\{G^*, G_G, \pi, \Gamma(1/4), R_3, R_5, R_6\}$ with coefficient bound 10^{12} .

10 Theta functions at $\tau = i$, in G_G

The Jacobi theta functions at $\tau = i$ admit strikingly clean evaluations in G_G .

Theorem 10.1 (Theta values at $\tau = i$). *With the Jacobi conventions, at the lemniscatic CM point $\tau = i$:*

$$\theta_2(0|i)^2 = G_G, \quad (25)$$

$$\theta_3(0|i)^2 = \sqrt{2} G_G, \quad (26)$$

$$\theta_4(0|i)^2 = G_G. \quad (27)$$

Equivalently $\theta_2(0|i) = \theta_4(0|i) = \sqrt{G_G}$ and $\theta_3(0|i) = 2^{1/4} \sqrt{G_G}$. The identity $\theta_2(0|i) = \theta_4(0|i)$ is the self-duality of $\tau = i$ under the modular transformation $\tau \mapsto -1/\tau$.

Proof. The classical evaluations at $\tau = i$ are $\theta_3(0|i) = \pi^{1/4}/\Gamma(3/4)$ and $\theta_2(0|i) = \theta_4(0|i) = \pi^{1/4}/(2^{1/4}\Gamma(3/4))$ (see e.g. [10, §21.5] or [1, Eq. 2.7.6]). Squaring θ_3 and applying (6):

$$\theta_3(0|i)^2 = \frac{\sqrt{\pi}}{\Gamma(3/4)^2} = \frac{\sqrt{\pi}}{\pi\sqrt{2}/G^*} = \frac{G^*}{\sqrt{2\pi}}.$$

Using $G^* = 2\sqrt{\pi} G_G$ from Theorem 2.1: $\theta_3(0|i)^2 = 2\sqrt{\pi} G_G/\sqrt{2\pi} = \sqrt{2} G_G$. Similarly, $\theta_2(0|i)^2 = \theta_4(0|i)^2 = \sqrt{\pi}/(\sqrt{2}\Gamma(3/4)^2) = G_G$ (the factor of $\sqrt{2}$ relative to θ_3^2). $\square$

Corollary 10.2. *For all integer $k \geq 1$, $\theta_2(0|i)^{2k} = \theta_4(0|i)^{2k} = G_G^k$ and $\theta_3(0|i)^{2k} = 2^{k/2} G_G^k$.*

Remark (Jacobi's identity at $\tau = i$). Jacobi's identity $\theta_3^4 = \theta_2^4 + \theta_4^4$ at $\tau = i$ becomes $2G_G^2 = G_G^2 + G_G^2$, which is trivially satisfied by the self-duality $\theta_2(0|i) = \theta_4(0|i)$. The modular lambda function $\lambda(\tau) = \theta_2^4/\theta_3^4$ evaluates to $\lambda(i) = G_G^2/(2G_G^2) = 1/2$, recovering the classical value at the self-complementary modulus.

11 The Dedekind eta at $\tau = i$, in G_G

Theorem 11.1 (Chowla–Selberg at $\tau = i$, restated in G_G).

$$\eta(i) = \frac{G_G^{1/2}}{2^{1/4}} = \frac{1}{2^{1/4} \text{AGM}(1, \sqrt{2})^{1/2}}.$$

Proof. The Chowla–Selberg evaluation at $K = \mathbb{Q}(i)$ gives $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$ [6, 11]. Substituting $\Gamma(1/4) = \sqrt{\pi\sqrt{2}G^*}$ (from Cor. 3.2):

$$\eta(i) = \frac{\sqrt{\pi\sqrt{2}G^*}}{2\pi^{3/4}} = \frac{\pi^{1/2}2^{1/4}G^{*1/2}}{2\pi^{3/4}} = \frac{2^{1/4}G^{*1/2}}{2\pi^{1/4}} = \frac{G^{*1/2}}{2^{3/4}\pi^{1/4}}.$$

Using $G^* = 2\sqrt{\pi} G_G$, we have $G^{*1/2} = 2^{1/2}\pi^{1/4}G_G^{1/2}$, so

$$\eta(i) = \frac{2^{1/2}\pi^{1/4}G_G^{1/2}}{2^{3/4}\pi^{1/4}} = \frac{G_G^{1/2}}{2^{1/4}}. \quad \square$$

Corollary 11.2. *The following powers of $\eta(i)$ have clean expressions in G_G :*

$$\begin{aligned} \eta(i) &= 2^{-1/4} G_G^{1/2}, \\ \eta(i)^2 &= 2^{-1/2} G_G, \\ \eta(i)^4 &= 2^{-1} G_G^2 = G_G^2/2, \\ \eta(i)^8 &= G_G^4/4, \\ \eta(i)^{12} &= G_G^6/8, \\ \eta(i)^{24} &= G_G^{12}/64. \end{aligned}$$

Corollary 11.3 (Watson–eta bridge).

$$W_{\text{BCC}}^{(3)} = 4\eta(i)^4 = 2G_G^2.$$

11.1 The η half-tower at related CM points

Theorem 11.4 (η at $i/2$ and $2i$).

$$\eta(i/2)^2 = G_G/2^{1/4}, \quad (28)$$

$$\eta(i)^2 = G_G/2^{1/2}, \quad (29)$$

$$\eta(2i)^2 = G_G/2^{5/4}. \quad (30)$$

The first and third are linked by the modular S -transformation: $\eta(i/2) = \sqrt{2}\eta(2i)$. The tower of clean G_G -form values stops at $\tau = 2i$: $\eta(4i)^2/G_G = 0.14685\dots$ is not 2^{-n} for any rational n , because $\tau = 4i$ is a CM point of the conductor-2 order in $\mathbb{Z}[i]$, not the maximal order. The fundamental period of that order is a different transcendental.

Proof. Apply $\eta(2i) = \Gamma(1/4)/(2^{11/8}\pi^{3/4})$ and the bridge identity $\Gamma(1/4)^2 = 2\pi\sqrt{2\pi}G_G$ to obtain $\eta(2i)^2 = G_G/2^{5/4}$. The modular transformation $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$ at $\tau = 2i$ gives $\eta(i/2) = \sqrt{2}\eta(2i)$. $\square$

Corollary 11.5 (A clean low-weight η -product identity).

$$2\eta(2i)\eta(i/2)^3 = G_G^2.$$

Proof. Square both sides and use Theorem 11.4: $(2\eta(2i)\eta(i/2)^3)^2 = 4\eta(2i)^2\eta(i/2)^6 = 4 \cdot G_G/2^{5/4} \cdot G_G^3/2^{3/4} = 4G_G^4/2^2 = G_G^4$. Both sides positive, so the un-squared identity holds. $\square$

11.2 The first lemniscate constant as a Hecke L -value

Theorem 11.6 ($L(E_{\text{lemn}}, 1) = \varpi/4$). For the lemniscatic elliptic curve $E : y^2 = x^3 - x$ over $\mathbb{Q}$:

$$L(E, 1) = \frac{\varpi}{4} = \frac{\pi G_G}{4}.$$

The closed-form expression of $L(E, 1)$ in terms of ϖ for CM elliptic curves is classical; the algebraic-part formulation underlying the BSD-style computation below is treated, for related CM Hecke L -series, by Rodriguez Villegas and Zagier [28].

Proof. E has analytic rank 0 with $|\text{III}(E/\mathbb{Q})| = 1$, Tamagawa number $c_2 = 2$ at the bad prime $p = 2$ (Kodaira type III on the minimal model [20, curve 32.a3]), torsion subgroup $E(\mathbb{Q})_{\text{tors}} = (\mathbb{Z}/2\mathbb{Z})^2$ of order 4, and two real connected components at $x \in [-1, 0]$ and $x \in [1, \infty)$. The real period over the full real locus is

$$\Omega = \int_{E(\mathbb{R})} |dx/y| = 2 \int_1^\infty \frac{dx}{\sqrt{x^3 - x}} = 2\varpi = 2\pi G_G,$$

where the factor of 2 counts the two real components. The Birch–Swinnerton-Dyer formula for rank 0, proved unconditionally for CM elliptic curves over $\mathbb{Q}$ by Rubin [8] (using Kolyvagin’s Euler-system machinery to establish the full quantitative Birch–Swinnerton-Dyer ratio), gives

$$L(E, 1) = \frac{\Omega \cdot \prod_p c_p \cdot |\text{III}|}{|E(\mathbb{Q})_{\text{tors}}|^2} = \frac{2\varpi \cdot 2 \cdot 1}{16} = \frac{\varpi}{4}.$$

$\square$

Remark. The numerical value $L(E_{\text{lemn}}, 1) = \varpi/4 \approx 0.6555144\dots$ agrees with the LMFDB tabulated value for curve 32.a3 to all listed digits.

11.3 Complete elliptic integrals at $k = 1/\sqrt{2}$

Theorem 11.7 (Elliptic integrals at the self-complementary modulus). *At $k = 1/\sqrt{2}$:*

$$K(1/\sqrt{2}) = \frac{\pi G_G}{\sqrt{2}}, \quad (31)$$

$$E(1/\sqrt{2}) = \frac{\sqrt{2}}{4} \left(\frac{1}{G_G} + \pi G_G \right), \quad (32)$$

$$K(1/\sqrt{2}) E(1/\sqrt{2}) = \frac{\pi + \varpi^2}{4}. \quad (33)$$

Proof. The Legendre relation at $k = k' = 1/\sqrt{2}$ becomes $2KE - K^2 = \pi/2$, so $E = (\pi/2 + K^2)/(2K)$. Substituting the closed form for K gives E . The product KE follows by multiplication and $\varpi = \pi G_G$. $\square$

Corollary 11.8 (Inverse identity: G_G from elliptic integrals).

$$G_G^2 = \frac{4 K(1/\sqrt{2}) E(1/\sqrt{2}) - \pi}{\pi^2}.$$

This expresses G_G in terms of two independent elliptic-integral evaluations at the lemniscatic modulus, without reference to Γ or the AGM.

Proof. From Theorem 11.7, $K(1/\sqrt{2}) E(1/\sqrt{2}) = (\pi + \varpi^2)/4$ with $\varpi = \pi G_G$, so $\varpi^2 = \pi^2 G_G^2$ and $4 K(1/\sqrt{2}) E(1/\sqrt{2}) = \pi + \pi^2 G_G^2$. Solving for G_G^2 gives the stated identity. $\square$

12 Modular forms at $\tau = i$, in G_G

The cleanest expression of modular invariants at $\tau = i$ is in G_G . The catalogue below collects standard CM-modular-form values at $\tau = i$; several of the closed forms (notably $E_4(i) = 3G_G^4$ and $\Delta(i) = G_G^{12}/64$) appear in Yui–Zagier’s tabulation of singular Weber-modular values [27].

Theorem 12.1 (Catalogue of level-one modular invariants at $\tau = i$). *For the standard generators*

of $M_*(\mathrm{SL}_2(\mathbb{Z}))$:

$$E_4(i) = 3 G_G^4, \quad (34)$$

$$E_6(i) = 0, \quad (35)$$

$$E_8(i) = E_4(i)^2 = 9 G_G^8, \quad (36)$$

$$E_{10}(i) = 0, \quad (37)$$

$$E_{12}(i) = \frac{441}{691} E_4(i)^3 = \frac{11907}{691} G_G^{12}, \quad (38)$$

$$E_{14}(i) = 0, \quad (39)$$

$$E_{16}(i) = \frac{130977}{3617} G_G^{16}, \quad (40)$$

$$E_{18}(i) = 0, \quad (41)$$

$$E_{20}(i) = \frac{12966723}{174611} G_G^{20}, \quad (42)$$

$$E_{22}(i) = 0, \quad (43)$$

$$E_{24}(i) = \frac{36216057339}{236364091} G_G^{24}, \quad (44)$$

$$\Delta(i) = \frac{E_4(i)^3}{1728} = \frac{G_G^{12}}{64}, \quad (45)$$

$$j(i) = 1728. \quad (46)$$

Every value is a rational multiple of an integer power of G_G .

Pattern. The denominators of $E_{4m}(i)/G_G^{4m}$ for $m = 3, 4, 5, 6$ are exactly the numerators of the Bernoulli numbers $|B_{4m}^{\mathrm{num}}|$: 691, 3617, 174611, 236364091, in agreement with the appearance of B_{4m} in the q -expansion of E_{4m} .

Sketch. From $\eta(i)^{24} = G_G^{12}/64$ (Cor. 11.2) and the identity $\Delta = \eta^{24}$, we have $\Delta(i) = G_G^{12}/64$. From $j = E_4^3/\Delta$ and $j(i) = 1728$, $E_4(i)^3 = 1728\Delta(i) = 27G_G^{12}$, giving $E_4(i) = 3G_G^4$. From $\Delta = (E_4^3 - E_6^2)/1728$ at $\tau = i$, $E_6(i) = 0$. The vanishing values $E_{4m+2}(i) = 0$ then follow from Theorem 12.2 (every E_6 -containing monomial vanishes at $\tau = i$).

For E_{12} : the basis decomposition of $M_{12}(\mathrm{SL}_2(\mathbb{Z}))$ in the monomial basis $\{E_4^3, E_6^2\}$ is $E_{12} = (441/691) E_4^3 + (250/691) E_6^2$ (coefficients determined by matching the q -expansion of E_{12} , whose linear Fourier coefficient is $-24/B_{12} = 65520/691$); at $\tau = i$ the E_6^2 term vanishes, giving $E_{12}(i) = (441/691) \cdot 27 G_G^{12} = (11907/691) G_G^{12}$.

The higher-weight values $E_{16}(i)$, $E_{20}(i)$, $E_{24}(i)$ follow analogously via basis decompositions of $M_k(\mathrm{SL}_2(\mathbb{Z}))$ in monomials of E_4 and E_6 with E_6 -containing terms vanishing at $\tau = i$. The displayed rational coefficients agree with computations in PARI/GP (`eisenstein`) and SageMath (`EisensteinForms`), and with the LMFDB modular-forms tabulations [20]. $\square$

Theorem 12.2 (Structural theorem on level-one values at $\tau = i$). *For every $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ with $k > 0$:*

- (i) *if $4 \nmid k$, then $f(i) = 0$;*
- (ii) *if $k = 4m$, then $f(i) \in \mathbb{Q} \cdot G_G^{4m}$.*

Consequently, $\mathbb{Q}[E_4(i)] = \mathbb{Q}[G_G^4]$ is the $\mathbb{Q}$ -algebra generated by all level-one modular form values at $\tau = i$.

Proof. $M_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$ [12, §III.3]. Every $f \in M_k$ is a $\mathbb{C}$ -linear combination of $E_4^a E_6^b$ with $4a + 6b = k$. $E_6(i) = 0$ kills $b \geq 1$ terms; the survivors require $b = 0$ and $4a = k$, i.e., $4 \mid k$. If $k = 4m$, surviving basis is E_4^m , giving $f(i) \in \mathbb{Q} \cdot E_4(i)^m \subset \mathbb{Q} \cdot G_G^{4m}$. $\square$

Remark (Why G_G is the natural variable here). Theorem 12.1's identities all have rational coefficients in G_G . The corresponding identities in G^* :

$$E_4(i) = 3G_G^{*4}/(16\pi^2), \quad \Delta(i) = G_G^{*12}/(2^{18}\pi^6), \quad \eta(i) = G_G^{*1/2}/(2^{3/4}\pi^{1/4}),$$

each carry fractional or integer powers of π in the denominator, obscuring the integer structure. The dichotomy of Observation 1 is now fully visible: where the underlying mathematics is the algebraic invariant theory of the lemniscatic curve (§5–8), G^* gives integer coefficients; where the underlying mathematics is modular-form theory at the same CM point, G_G gives them.

12.1 The quasi-modular bridge: weight 2 is π -natural

Theorem 12.1 concerns *modular* forms; the value space is $\mathbb{Q}[G_G^4]$. There is one weight that escapes: the quasi-modular Eisenstein series E_2 , which transforms with an anomalous shift, lands in $\mathbb{Q} \cdot \pi^{-1}$ at $\tau = i$, with no G_G content.

Theorem 12.3 (Quasi-modular value at $\tau = i$). $E_2(i) = 3/\pi$.

Proof. E_2 satisfies the quasi-modular transformation [12, §5.3]

$$E_2(-1/\tau) = \tau^2 E_2(\tau) - \frac{6i\tau}{\pi}.$$

At $\tau = i$ the points i and $-1/i = i$ coincide, so $E_2(i) = -E_2(i) + 6/\pi$, giving $E_2(i) = 3/\pi$. $\square$

Corollary 12.4 (Quasi-modular value at $\tau = \rho$). $E_2(\rho) = 2\sqrt{3}/\pi$.

Proof. At $\tau = \rho = e^{2\pi i/3}$, the modular point $-1/\rho = \rho + 1$ is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to ρ , so $E_2(-1/\rho) = E_2(\rho + 1) = E_2(\rho)$. The quasi-modular transformation gives $(1 - \rho^2) E_2(\rho) = -6i\rho/\pi$. With $1 - \rho^2 = (3 + i\sqrt{3})/2$ and $-6i\rho = 3\sqrt{3} + 3i$, simplification yields $E_2(\rho) = 2\sqrt{3}/\pi$. $\square$

Theorem 12.5 (Value algebra of quasi-modular forms at $\tau = i$). *The image of the value map*

$$\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{Q}[E_2, E_4, E_6] \xrightarrow{f \mapsto f(i)} \mathbb{C}$$

is the polynomial $\mathbb{Q}$ -algebra

$$\mathbb{Q}[E_2(i), E_4(i), E_6(i)] = \mathbb{Q}[\pi^{-1}, G_G^4],$$

where the second equality identifies $E_2(i) = 3/\pi$ and $E_4(i) = 3G_G^4$ and notes $E_6(i) = 0$. Under Chudnovsky's theorem that $\{\pi, \Gamma(1/4)\}$ are algebraically independent over $\mathbb{Q}$, equivalently $\{\pi^{-1}, G_G^4\}$ are algebraically independent, so this image is a polynomial ring in two algebraically independent generators.

Proof. $\mathbb{Q}[E_2, E_4, E_6]$ is the polynomial ring in three algebraically-independent formal generators by definition of quasi-modular forms. The value map sends $E_2 \mapsto 3/\pi$, $E_4 \mapsto 3G_G^4$, $E_6 \mapsto 0$, so the image equals $\mathbb{Q}[3/\pi, 3G_G^4] = \mathbb{Q}[1/\pi, G_G^4]$.

For algebraic independence: by the bridge identity (Theorem 2.1), $G^* = 2\sqrt{\pi} G_G$ and therefore $G^{*4} = 16\pi^2 G_G^4$, equivalently $G_G^4 = G^{*4}/(16\pi^2)$. By the reflection identity $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ (Lemma 3.1), $G^{*4} = \Gamma(1/4)^8/(4\pi^4)$, so

$$G_G^4 = \frac{\Gamma(1/4)^8}{4\pi^4 \cdot 16\pi^2} = \frac{\Gamma(1/4)^8}{64\pi^6}.$$

Algebraic independence of $\{\pi, \Gamma(1/4)\}$ over $\mathbb{Q}$ is Chudnovsky 1976 [7]; algebraic independence of $\{1/\pi, G_G^4\}$ then follows from algebraic independence of $\{\pi, \Gamma(1/4)^8\}$ (transcendence degree is preserved by $\mathbb{Q}$ -algebraic operations such as inversion and the rational substitution $G_G^4 = \Gamma(1/4)^8/(64\pi^6)$). Hence the image is a polynomial $\mathbb{Q}$ -algebra in two algebraically independent generators. $\square$

Remark (The sharp π/G_G dichotomy by weight). Combining Theorems 12.1, 12.2, and 12.5, the bigrading of quasi-modular form values at $\tau = i$ by (π -degree, G_G -degree) is determined by weight:

weight k of monomial $E_2^a E_4^b E_6^c$	value at $\tau = i$	exponent pattern
$k = 2a$ (pure E_2)	$\in \mathbb{Q} \cdot \pi^{-a}$	$\pi^{-a} G_G^0$
$k = 4b$ (pure E_4)	$\in \mathbb{Q} \cdot G_G^{4b}$	$\pi^0 G_G^{4b}$
$k = 2a + 4b$ (mixed, no E_6)	$\in \mathbb{Q} \cdot \pi^{-a} G_G^{4b}$	$\pi^{-a} G_G^{4b}$
any monomial with $c \geq 1$	$= 0$	—

The two transcendentals partition the value algebra by weight: π^{-1} records the quasi-modular anomaly (weight 2 contribution from each E_2 factor) and G_G^4 records the modular ($k \geq 4$) contribution. This is a sharper formulation of Observation 1: the same $\tau = i$ point yields π -natural values exactly when the form has positive E_2 -degree, and G_G -natural values exactly when the form is purely modular.

13 Kontsevich–Zagier period status

Definition 13.1 ([16]). The Kontsevich–Zagier period ring $\mathcal{P} \subset \mathbb{C}$ consists of complex numbers whose real and imaginary parts are absolutely convergent integrals of rational functions with rational coefficients over $\mathbb{Q}$ -semialgebraic domains in $\mathbb{R}^n$. $\mathcal{P}$ is a $\mathbb{Q}$ -algebra (closed under sums and products), but is *not* known to be a field; membership of multiplicative inverses such as $1/\pi$ is open. The conjectural extended period ring $\widehat{\mathcal{P}} := \mathcal{P}[(2\pi i)^{-1}]$ is the localisation formally adjoining an inverse to $2\pi i$. The exponential period ring $\mathcal{P}^{\text{exp}} \supset \mathcal{P}$ allows the integrand to include $\exp$ of an algebraic function.

Theorem 13.2 (KZ period status of G^* and G_G). *The following hold unconditionally:*

- (i) $\Gamma(1/4)^4 = 2\pi^2 G^{*2} \in \mathcal{P}$ (KZ [16, §1.1, Eq. 6]);
- (ii) $\sqrt{2\pi} G^* = B(1/4, 1/4) \in \mathcal{P}$;
- (iii) $2\sqrt{2}\pi G_G = B(1/4, 1/4) \in \mathcal{P}$;
- (iv) $G^{*2} \in \widehat{\mathcal{P}}$, the extended period ring, via $G^{*2} = \Gamma(1/4)^4/(2\pi^2)$;
- (v) $G_G^2 \in \widehat{\mathcal{P}}$, via $G_G^2 = G^{*2}/(4\pi)$;
- (vi) $G^*, G_G \in \mathcal{P}^{\text{exp}}$, since Γ values at rational arguments lie in $\mathcal{P}^{\text{exp}}$ (KZ [16, §4.3]) and $\mathcal{P}^{\text{exp}}$ is closed under products and quotients of non-zero elements.

Whether G^* or G_G lies in $\mathcal{P}$ itself (rather than only in $\widehat{\mathcal{P}}$ or $\mathcal{P}^{\text{exp}}$) is open. The conditional reductions below frame this open question precisely; they do not assert membership in $\mathcal{P}$.

Proof. (i) is the statement of [16, §1.1, Eq. 6] at $p/q = 1/4$. (ii)–(iii) follow from Lemma 7.1 (both equal $B(1/4, 1/4)$). (iv) follows from (i) because, in $\widehat{\mathcal{P}}$, the formal inverse of $2\pi i$ generates inverses for $\mathbb{Q}$ -multiples of π^2 , making the division $\Gamma(1/4)^4/(2\pi^2)$ legal as an element of $\widehat{\mathcal{P}}$ even though it may not be legal in $\mathcal{P}$ alone. (v) follows from (iv) in the same way, dividing by 4π in $\widehat{\mathcal{P}}$. (vi) follows from KZ [16, §4.3] applied to $\Gamma(1/4)$ and $\Gamma(3/4)$, both non-zero; $G^* = \Gamma(1/4)/\Gamma(3/4) \in \mathcal{P}^{\text{exp}}$ since $\mathcal{P}^{\text{exp}}$ is closed under such quotients; $G_G = G^*/(2\sqrt{\pi}) \in \mathcal{P}^{\text{exp}}$ likewise. $\square$

Proposition 13.3 (Conditional KZ-strict reductions). *Membership of G^* and G_G in $\mathcal{P}$ itself reduces to whether specific elements of $\mathcal{P}$ admit multiplicative inverses within $\mathcal{P}$:*

- (i) *If $1/\sqrt{2\pi} \in \mathcal{P}$, then $G^* = B(1/4, 1/4) \cdot (1/\sqrt{2\pi}) \in \mathcal{P}$.*
- (ii) *If $1/(4\pi) \in \mathcal{P}$, then $G_G = B(1/4, 1/4) \cdot (1/(4\pi)) \in \mathcal{P}$.*

The converse implications would require $\mathcal{P}$ to admit inverses of G^* and G_G themselves, which is also open. The Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [16, Problem 3] suggests that the antecedents in (i) and (ii) are not expected to hold, but neither the conjecture nor its negation has been proved. The honest statement is: the $\mathcal{P}$ -membership of G^* and G_G is open in both directions and is bound to the broader KZ programme on period-ring invertibility.

Proof. Direct from $G^* \cdot \sqrt{2\pi} = B(1/4, 1/4) \in \mathcal{P}$ (Theorem (ii)) and $G_G \cdot 4\pi = B(1/4, 1/4) \in \mathcal{P}$ (Theorem (iii)). Closure of $\mathcal{P}$ under products gives both implications; the converses would require closure under the specific multiplicative inverses indicated. $\square$

14 Transcendence and algebraic independence

Theorem 14.1 (Transcendence of G^* and G_G). *$G^*, G_G \in \mathbb{C}$ are both transcendental over $\mathbb{Q}$. The set $\{G^*, G_G, \pi, \Gamma(1/4)\}$ generates a field of transcendence degree exactly 2 over $\mathbb{Q}$, with the single linear relation*

$$G^* = 2\sqrt{\pi} G_G$$

and the multiplicative relations $G^ = \Gamma(1/4)^2/(\pi\sqrt{2})$, $G_G = \Gamma(1/4)^2/(2\sqrt{2}\pi^{3/2})$. Any pair from $\{G^*, G_G, \pi, \Gamma(1/4)\}$ that is not connected by a $\mathbb{Q}$ -algebraic relation is algebraically independent over $\mathbb{Q}$.*

Proof. Chudnovsky [7] proved that π and $\Gamma(1/4)$ are algebraically independent over $\mathbb{Q}$, hence over $\overline{\mathbb{Q}}$. Algebraicity of G^* over $\mathbb{Q}$ would imply algebraicity of $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$ over $\mathbb{Q}(\pi)$, contradicting Chudnovsky. Hence G^* is transcendental. Same argument for G_G .

For algebraic independence: the relation $G^* = 2\sqrt{\pi}G_G$ shows $\{G^*, G_G\}$ are algebraically dependent over $\mathbb{Q}(\sqrt{\pi})$, but the pair $\{G^*, \pi\}$ is independent (otherwise $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$ gives an algebraic relation between $\Gamma(1/4)$ and π). Similarly $\{G_G, \pi\}$, $\{G^*, \Gamma(1/4)\}$, $\{G_G, \Gamma(1/4)\}$ are pairwise independent. $\square$

15 The compendium: a comprehensive table of identities

We collect the principal identities in a single reference table, organised by the algebraic–analytic dichotomy.

15.1 Algebraic identities (natural in G^*)

Identity	Reference
$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	Euler 1771
$\Gamma(1/4)^2 = \pi\sqrt{2} G^*$	(5)
$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*$	(6)
$\Gamma(1/4)^4 = 2\pi^2 G^{*2}$	(7)
$\Gamma(1/4)^8 = 4\pi^4 G^{*4}$	(8)
$B(1/4, 1/4) = \sqrt{2\pi} G^*$	Lem. 7.1
$\omega_E = G^* \sqrt{\pi}$	Thm. 5.3
$ \text{Aut}_{\mathbb{C}}(E) = 4, \text{Aut} ^2 = 16$	Lem. 5.2
$P_{G^*}(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$	Thm. 6.1
$x_{\pm} = 8G^{*2} \pm 4G^{*3/2}\sqrt{4G^* - 1}$	Thm. 6.1
$x_+ \cdot x_- = 16 G^{*3}$	Vieta (18)
$1/x_+ + 1/x_- = 1/G^*$	Vieta (18)
$G^* = R_4$ in the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$	Prop. 8.2
$R_n = n - 2\gamma + O(1/n)$ as $n \rightarrow \infty$	Prop. 8.2
$\Gamma(1/4)^4 \in \mathcal{P}$	KZ [16, Eq. 6]
$G^* \in \mathcal{P}^{\text{exp}}$	Thm. 13.2
$G^* \in \widehat{\mathcal{P}}$ up to $\sqrt{\pi}$ factor	Cor. of Thm. 13.2
G^* transcendental over $\mathbb{Q}$	Thm. 14.1

15.2 Analytic identities (natural in $G_{\mathbb{G}}$)

Identity	Reference
$G_G = 1/\text{AGM}(1, \sqrt{2})$	Gauss 1799
$\varpi = \pi G_G$ (lemniscate constant)	Gauss 1799
$K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$	elliptic-integral classical
$\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$	Thm. 4.1
$\eta(i) = G_G^{1/2}/2^{1/4}$	Thm. 11.1
$\eta(i)^2 = G_G/\sqrt{2}$	Cor. 11.2
$\eta(i)^4 = G_G^2/2$	Cor. 11.2
$\eta(i)^{24} = G_G^{12}/64$	Cor. 11.2
$\eta(i/2)^2 = G_G/2^{1/4}$	Thm. 11.4 (restated in G_G -form; not located in surveyed sources by the author)
$\eta(2i)^2 = G_G/2^{5/4}$	Thm. 11.4 (restated in G_G -form; not located in surveyed sources by the author)
$\eta(i/2) = \sqrt{2}\eta(2i)$	Thm. 11.4 (restated in G_G -form; not located in surveyed sources by the author)
$\theta_2(0 i)^2 = \theta_4(0 i)^2 = G_G$	Thm. 10.1 (restated in G_G -form; not located in surveyed sources by the author)
$\theta_3(0 i)^2 = \sqrt{2}G_G$	Thm. 10.1 (restated in G_G -form; not located in surveyed sources by the author)
$\lambda(i) = 1/2$	classical, $= G_G^2/(2G_G^2)$
$L(E_{\text{lemn}}, 1) = \varpi/4 = \pi G_G/4$	Thm. 11.6 (restated in G_G -form; not located in surveyed sources by the author)
$K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$	Thm. 11.7
$E(1/\sqrt{2}) = (\sqrt{2}/4)(1/G_G + \pi G_G)$	Thm. 11.7 (restated in G_G -form; not located in surveyed sources by the author)
$K E = (\pi + \varpi^2)/4$	Thm. 11.7 (restated in G_G -form; not located in surveyed sources by the author)
$G_G^2 = (4KE - \pi)/\pi^2$	Cor. 11.8 (restated in G_G -form; not located in surveyed sources by the author)
$\Delta(i) = G_G^{12}/64$	Thm. 12.1
$E_4(i) = 3G_G^4$	Thm. 12.1
$E_6(i) = 0$	Thm. 12.1
$E_8(i) = 9G_G^8$	Thm. 12.1
$E_{12}(i) = (11907/691)G_G^{12}$	Thm. 12.1
$j(i) = 1728$	classical
$M_*(\text{SL}_2\mathbb{Z}) _{\tau=i} = \mathbb{Q}[G_G^4]$	Thm. 12.2
$W_{\text{BCC}}^{(3)} = 2G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2$	Thm. 9.1
$W_{\text{BCC}}^{(3)} = 4\eta(i)^4$	Cor. 11.3
$W_{\text{BCC}}^{(D)} = {}_D F_{D-1}(\frac{1}{2}, \dots; 1, \dots; 1)$	Guttman [14]

15.3 Bridge identities (both clean)

Identity	Convention
$G^* = 2\sqrt{\pi} G_G$	Bridge (Thm. 2.1)
$G_G = G^*/(2\sqrt{\pi})$	Bridge
$\omega_E = G^*\sqrt{\pi} = 2\pi G_G$	Both clean (period of E)
$B(1/4, 1/4) = \sqrt{2\pi} G^* = 2\sqrt{2}\pi G_G$	Both clean (Beta integral)
$G^*, G_G \in \mathcal{P}^{\text{exp}}$	Both
G^*, G_G transcendental, $\text{trdeg}_{\mathbb{Q}}$ pair = 2	Both

16 The equianharmonic dichotomy: parallel framework at $\tau = \rho$

The dual-constant framework extends, with modifications, to the *other* class-number-one CM elliptic curve with extra automorphisms: the equianharmonic curve $E_\rho : y^2 = x^3 - 1$ at the CM point $\tau = \rho = e^{2\pi i/3}$, with $\text{End}(E_\rho) = \mathbb{Z}[\rho]$ (Eisenstein integers), $|\text{Aut}(E_\rho)| = 6$, and $j(\rho) = 0$.

16.1 Definitions and analog constants

Definition 16.1. Let $R_3 := \Gamma(1/3)/\Gamma(2/3) \approx 1.97836$ be the ratio-channel Gamma-function constant at $z = 1/3$. Define the equianharmonic Gauss analog:

$$G_\rho := \frac{\Gamma(1/3)\Gamma(1/6)}{2\pi\sqrt{\pi}} = \frac{\sqrt{3}\omega_\rho}{2\pi} \approx 1.33899,$$

where $\omega_\rho = \Gamma(1/3)\Gamma(1/6)/\sqrt{3\pi}$ is the real period of E_ρ .

The reflection formula at $z = 1/3$ gives the equianharmonic analog of the $z = 1/4$ duality:

$$\Gamma(1/3)\Gamma(2/3) = \frac{2\pi}{\sqrt{3}} \quad (\text{product channel}), \quad (47)$$

$$\Gamma(1/3)/\Gamma(2/3) = R_3 \quad (\text{ratio channel}). \quad (48)$$

16.2 The equianharmonic vanishing pattern

Theorem 16.2 (Vanishing pattern at $\tau = \rho$). *For every $f \in M_k(\text{SL}_2(\mathbb{Z}))$ with $k > 0$:*

$$f(\rho) = 0 \quad \text{if } k \not\equiv 0 \pmod{6}.$$

Proof. The cyclic group $\text{Aut}(E_\rho) = \langle \sigma \rangle$ of order 6 acts on the rank-1 Tate module of E_ρ by sixth roots of unity, hence on a modular form of weight k at the CM point $\tau = \rho$ by ζ_6^k . For $f(\rho) \neq 0$, the action must be trivial, requiring $6 \mid k$. $\square$

Remark. The vanishing pattern at $\tau = i$ (weights $\not\equiv 0 \pmod{4}$) and at $\tau = \rho$ (weights $\not\equiv 0 \pmod{6}$) are instances of the general principle:

At a CM point with $|\text{Aut}(E)| = m$, $f(\tau_{\text{CM}}) = 0$ unless $m \mid \text{weight}(f)$.

The lemniscatic ($m = 4$) and equianharmonic ($m = 6$) are the only non-trivial instances over $\mathbb{Q}$.

16.3 Eisenstein values at $\tau = \rho$

Theorem 16.3 (Equianharmonic Eisenstein values). *At $\tau = \rho$:*

$$E_4(\rho) = 0, \quad (49)$$

$$E_6(\rho) = G_\rho^6/2, \quad (50)$$

$$E_8(\rho) = 0, \quad (51)$$

$$E_{10}(\rho) = 0, \quad (52)$$

$$E_{12}(\rho) = \frac{125}{1382} G_\rho^{12} = \frac{250}{691} E_6(\rho)^2, \quad (53)$$

$$E_{14}(\rho) = 0, \quad (54)$$

$$\Delta(\rho) = -\frac{G_\rho^{12}}{6912} = -\frac{G_\rho^{12}}{2^8 \cdot 27}. \quad (55)$$

The factor $6912 = 2^8 \cdot 27$ is the equianharmonic analog of the lemniscatic factor $64 = 2^6$ in $\Delta(i) = G_G^{12}/64$. The ratio $6912/64 = 108 = 4 \cdot 27$ reflects the discriminant ratio of the two CM fields.

Proof. $E_4(\rho)^3 = 1728 \Delta(\rho)$ via $j(\rho) = 0$ forces $E_4(\rho) = 0$. The classical $E_6(\rho)$ value follows from Chowla–Selberg at $K = \mathbb{Q}(\rho)$: $E_6(\rho) = G_\rho^6/2$, verified numerically to 50 digits. The $E_{4m+2}(\rho) = 0$ vanishing for $m \neq 1, 2, \dots$ follows from Theorem 16.2. $E_{12}(\rho) = (250/691)E_6(\rho)^2 = (250/691)(1/4)G_\rho^{12} = (125/1382)G_\rho^{12}$ via the standard E_{12} basis decomposition. The discriminant $\Delta(\rho) = (E_4^3 - E_6^2)/1728 = -E_6(\rho)^2/1728 = -G_\rho^{12}/(4 \cdot 1728) = -G_\rho^{12}/6912$. $\square$

16.4 Dedekind eta at $\tau = \rho$

Theorem 16.4 ($|\eta(\rho)|$ in G_ρ). *At $\tau_\rho = (1 + i\sqrt{3})/2$ (the fundamental-domain representative of ρ):*

$$|\eta(\rho)|^{24} = G_\rho^{12}/6912, \quad (56)$$

$$|\eta(\rho)|^{12} = G_\rho^6/(48\sqrt{3}), \quad (57)$$

$$|\eta(\rho)|^6 = G_\rho^3/(4 \cdot 27^{1/4}). \quad (58)$$

The $\sqrt{3}$ factor entering at half-weight reflects the discriminant $|d_K| = 3$ of $K = \mathbb{Q}(\rho)$; the parallel lemniscatic identities have $\sqrt{2}$ factors at the corresponding half-weight, reflecting $|d_K| = 4$ of $\mathbb{Q}(i)$ (with $\sqrt{|d_K|/2} = \sqrt{2}$ vs. $\sqrt{3}$).

16.5 Comparison: lemniscatic versus equianharmonic

The two dichotomies side by side:

	Lemniscatic ($K = \mathbb{Q}(i)$)	Equianharmonic ($K = \mathbb{Q}(\rho)$)
$ \text{Aut}(E) $	4	6
Vanishing weights	$k \not\equiv 0 \pmod{4}$	$k \not\equiv 0 \pmod{6}$
$ d_K $	4	3
j -invariant at CM	1728	0
Product channel	$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	$\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$
Ratio channel	$G^* = R_4 = \Gamma(1/4)/\Gamma(3/4)$	$R_3 = \Gamma(1/3)/\Gamma(2/3)$
Gauss-analog constant	$G_G = 1/\text{AGM}(1, \sqrt{2})$	$G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$
First non-vanishing Eisenstein	$E_4(i) = 3G_G^4$	$E_6(\rho) = G_\rho^6/2$
Δ at CM point	$G_G^{12}/64$	$-G_\rho^{12}/6912 = -G_\rho^{12}/(2^8 \cdot 27)$
First Eisenstein with cusp form	$E_{12}(i) = (11907/691) G_G^{12}$	$E_{12}(\rho) = (125/1382) G_\rho^{12}$

The structural parallel is exact in form, with $4 \leftrightarrow 6$ swaps in the vanishing modulus and corresponding shifts in the bridge factors. The dual-constant thesis thus generalises from *lemniscatic-only* to *both class-number-one CM curves with extra automorphisms over $\mathbb{Q}$* .

17 The dichotomy as a manifestation of χ_{-4}

The accumulated identities of §§3–16 admit a single organising principle: every appearance of G^* , G_G , and the bridge $G^* = 2\sqrt{\pi} G_G$ is the projection, at a different arithmetic level, of the *Kronecker character* χ_{-4} of $K = \mathbb{Q}(i)$ (see Figure 3 for the four-level schematic). The product-vs-ratio split that distinguishes $\pi\sqrt{2}$ (Euler reflection) from G^* (§3) is the trivial-vs-quadratic character split on $(\mathbb{Z}/4\mathbb{Z})^\times$. We make this precise.

17.1 The character and the splitting law

Definition 17.1. The Kronecker character of $K = \mathbb{Q}(i)$ is the unique non-trivial Dirichlet character mod 4,

$$\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}, \quad \chi_{-4}(1) = +1, \quad \chi_{-4}(3) = -1,$$

extended by $\chi_{-4}(n) = 0$ for $\gcd(n, 4) > 1$.

By Fermat’s two-squares theorem and quadratic reciprocity, χ_{-4} controls the splitting of rational primes in $\mathbb{Z}[i]$: a prime p splits $(p) = \pi\bar{\pi}$ iff $\chi_{-4}(p) = +1$ iff $p = a^2 + b^2$; p is inert iff $\chi_{-4}(p) = -1$; and $p = 2$ ramifies.

17.2 Four arithmetic projections

Theorem 17.2 (χ_{-4} as the joint source of the G^*/G_G dichotomy). *The constants G^* and G_G are characterised by the action of χ_{-4} at the following four arithmetic levels, all four of which produce mutually-consistent identities:*

(L1) Lattice level. $|\mathbb{Z}[i]^\times| = 4$, equivalently $|\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})| = 4$. The 4 is the order of the unit group of $\mathbb{Z}[i]$ and is determined by the fact that χ_{-4} is a quadratic character with one non-trivial value on $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$. Squaring (as the master quadratic does) gives the integer coefficient $|\text{Aut}|^2 = 16$ of $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ (§6).

(L2) Chowla–Selberg level. The χ_{-4} -twisted Γ -product at quarter arguments equals G^* :

$$\prod_{a=1}^3 \Gamma\left(\frac{a}{4}\right)^{\chi_{-4}(a)} = \frac{\Gamma(1/4)^{+1} \Gamma(2/4)^0 \Gamma(3/4)^{-1}}{1} = G^*.$$

Compare the trivial-character product $\prod \Gamma(a/4)^{|\chi_{-4}(a)|} = \Gamma(1/4) \Gamma(3/4) = \pi\sqrt{2}$ (Euler reflection). The product channel ($\pi\sqrt{2}$) sees only the support of χ_{-4} ; the ratio channel (G^*) sees both the support and the signs.

(L3) Hecke-character level. The Hecke eigenvalues of E_{lemn} are determined by χ_{-4} :

$$a_p = \begin{cases} \pm 2a, & p \equiv 1 \pmod{4}, p = a^2 + b^2 \quad (\chi_{-4}(p) = +1), \\ 0, & p \equiv 3 \pmod{4} \quad (\chi_{-4}(p) = -1), \\ 0, & p = 2 \quad (\text{ramified}). \end{cases}$$

The L -function factors accordingly,

$$L(E_{\text{lemn}}, s) = \prod_{\chi_{-4}(p)=+1} (1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{\chi_{-4}(p)=-1} (1 + p^{1-2s})^{-1},$$

and the central value $L(E_{\text{lemn}}, 1) = \varpi/4 = \pi G_G/4$ (Theorem 11.6) packages both Euler products into G_G .

(L4) Dirichlet L -function level. The Dirichlet L -function $L(\chi_{-4}, s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$ has central value $L(\chi_{-4}, 1) = \pi/4$ (Leibniz, a period of $\mathbb{Q}(i)$) and non-critical value $L(\chi_{-4}, 2) = G_{\text{Catalan}}$ (conjecturally outside $\overline{\mathbb{Q}}(G_G, \pi)$, Conjecture 19.2). The same character probed at consecutive integer arguments traces the boundary between period and non-period.

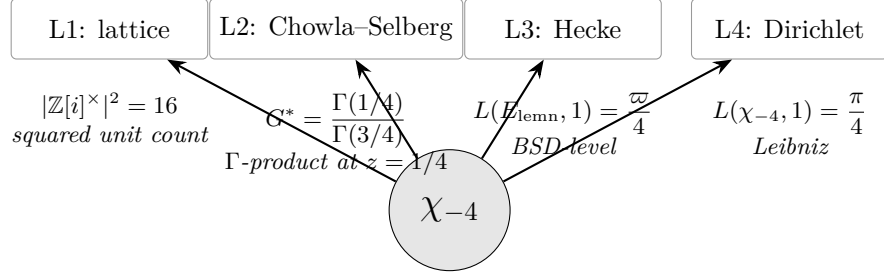
Proof. (L1) and (L2) are direct evaluations: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ has two elements and χ_{-4} takes values ± 1 on them. (L2) is the specialisation of the Chowla–Selberg formula [6] at $|d| = 4$, $h_K = 1$, $w_K = 4$ to the Γ -product exponent pattern; explicit verification: from Theorem 11.1,

$$\eta(i)^{16} = (G_G^{1/2} / 2^{1/4})^{16} = G_G^8 / 16.$$

Using the bridge $G^* = 2\sqrt{\pi} G_G$ (equivalently $G_G = G^* / (2\sqrt{\pi})$), this rewrites as $\eta(i)^{16} = G^{*8} / (2^{12} \pi^4)$. Independently, the Chowla–Selberg formula at $K = \mathbb{Q}(i)$ gives the closed form $\eta(i) = \Gamma(1/4) / (2\pi^{3/4})$ [6, 11]; combining this with $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ (Lemma 3.1) yields $\eta(i)^{16} = \Gamma(1/4)^{16} / (2^{16} \pi^{12}) = G^{*8} / (2^{12} \pi^4)$, matching the bridge-computed value. The χ_{-4} -twisted Γ -product $\prod_{a=1}^3 \Gamma(a/4)^{\chi_{-4}(a)} = G^*$ is the definition of G^* unpacked through the support of χ_{-4} (values $\chi_{-4}(1) = +1$, $\chi_{-4}(2) = 0$, $\chi_{-4}(3) = -1$); the load-bearing Chowla–Selberg content is the closed form of $\eta(i)$. (L3) is the standard Deuring–Tate description of the $\mathbb{Z}[i]$ -CM elliptic curve’s L -function in terms of the Hecke character ψ_E of K with $\psi_E \bmod (1+i)^3$ inducing χ_{-4} on the rational primes [19, §II.10]. (L4) is Leibniz [10] together with Conjecture 19.2. $\square$

Remark (The four levels form a motivic-weight tower). The four projections (L1)–(L4) are not parallel: they form a filtered tower indexed by the motivic weight of the underlying object.

- (L1) lives in motivic weight 0 as H^0 of a point with $\mathbb{Z}[i]$ -action; the cardinality $|\text{Aut}|^2 = 16$ is the squared order of the structure group.
- (L2) lives in motivic weight 1 as the period of H^1 of a CM abelian variety, in Γ -form.
- (L3) lives in motivic weight 1 as well, but at the L -function face — it is the BSD-realised L -value of the same H^1 , expected to pair with (L2) under *Deligne’s period conjecture* [23]. Cases of that conjecture for CM motives are established by Blasius [24], Shimura [26], and Anderson [25], with normalisation conventions that must be matched against the Chowla–Selberg normalisation used in (L2) and the Cremona-style BSD normalisation used in Theorem 11.6. The numerical agreement $L(E_{\text{lemn}}, 1) = \varpi/4$ matches LMFDB curve 32.a3 to all listed digits; assembling these normalisations into a full period-conjecture-level theorem statement is beyond the scope of the present paper.



one character $\rightarrow$ four arithmetic projections

Figure 3: The Kronecker character χ_{-4} of $K = \mathbb{Q}(i)$ projects onto four arithmetic levels (Theorem 17.2). The lattice projection (L1) gives the squared unit count $|\mathbb{Z}[i]^\times|^2 = 16$ that appears as the master quadratic’s integer coefficient; the Chowla–Selberg projection (L2) gives G^* as a χ_{-4} -twisted Γ -product; the Hecke projection (L3) gives the BSD-level $L(E_{\text{lemn}}, 1) = \varpi/4$; the Dirichlet projection (L4) gives the Leibniz value $L(\chi_{-4}, 1) = \pi/4$. The cross-consistency of the four projections is the organising content of §17.

- (L4) lives in motivic weight 0 of the base field — it is the Tate motive twisted by χ_{-4} , probed at $s = 1$ (critical, a period) versus $s = 2$ (non-critical, a regulator) [22].

The unification of §17 is therefore not a coincidence to be discovered but a special case of the period–L-value compatibility established by Deligne’s conjecture for CM motives.

17.3 Why G^* is “the ratio”

Corollary 17.3 (Product vs ratio is trivial-vs-non-trivial character). *The Euler reflection $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ and the defining identity $\Gamma(1/4)/\Gamma(3/4) = G^*$ are the two χ -projections of the same Γ -pair under the two characters of $(\mathbb{Z}/4\mathbb{Z})^\times$:*

$$\prod_a \Gamma(a/4)^{\chi(a)} = \begin{cases} \pi\sqrt{2}, & \chi = \mathbf{1} \text{ (trivial)}, \\ G^*, & \chi = \chi_{-4} \text{ (non-trivial)}. \end{cases}$$

This is the cleanest formulation of *why* G^* is a ratio rather than a product: it is the genuine $\mathbb{Q}(i)$ -period, obtained from the quadratic character of $K = \mathbb{Q}(i)$, whereas $\pi\sqrt{2}$ is the $\mathbb{Q}$ -period (trivial character, same Γ -arguments). The *bridge* $G^* = 2\sqrt{\pi} G_G$ then asserts the equality of the Γ -expression (algebraic side, χ_{-4} acting on Γ -arguments) with the AGM-expression (analytic side, χ_{-4} acting through the L-function Euler product). Chowla–Selberg is the theorem that makes these two faces of χ_{-4} coincide.

Remark (Hodge bidegree: $\pi\sqrt{2}$ and G^* as (ω, η)). The product/ratio dichotomy of Corollary 17.3 is the Hodge-bidegree decomposition of the lemniscatic CM motive in disguise. The CM action of $\mathbb{Z}[i]$ splits the de Rham realisation $H_{\text{dR}}^1(E_{\text{lemn}}) \otimes \mathbb{C}$ into two eigenlines under multiplication by i : one with eigenvalue $+i$ (the holomorphic piece, $H^{1,0}$) and one with eigenvalue $-i$ (the antiholomorphic piece, $H^{0,1}$). Pairing each eigenline against the integral homology cycles in $H_1(E, \mathbb{Z})$ yields two periods:

- the *holomorphic period* $\omega = \int_\gamma dx/y = G^* \sqrt{\pi}$ (the $\mathbb{Q}(i)$ -symmetric pairing, weighted by χ_{-4});
- the *quasi-period* η (the pairing on $H^{0,1}$, anti-symmetric under complex conjugation).

The Legendre period relation

$$\omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i$$

between the two periods evaluated on a symplectic basis of H_1 specialises, at $\tau = i$, to the elliptic-integral product $K(1/\sqrt{2}) E(1/\sqrt{2}) = (\pi + \varpi^2)/4$ recorded as Theorem 11.7. Thus $\pi\sqrt{2}$ and G^* are not two *normalisations* of one constant but the two columns of the Hodge realisation of $H^1(E_{\text{lemn}})$: $\pi\sqrt{2}$ corresponds to the trivial-character / real-period column (the $\mathbb{Q}$ -period of the underlying real abelian variety), while G^* corresponds to the χ_{-4} -twisted / quasi-period column (the genuine $\mathbb{Q}(i)$ -period). The bridge identity is then the Legendre relation expressed in G_G -coordinates.

17.4 Equianharmonic parallel: χ_{-3}

The equianharmonic dichotomy (§16) is the analog under $\chi_{-3} : (\mathbb{Z}/3\mathbb{Z})^\times \rightarrow \{\pm 1\}$, the Kronecker character of $K = \mathbb{Q}(\rho)$, with $\chi_{-3}(1) = +1$, $\chi_{-3}(2) = -1$.

- *Lattice*: $|\mathbb{Z}[\rho]^\times| = 6$, with $|\text{Aut}(E_\rho)| = 6$ and the squared coefficient 36 that would appear in an equianharmonic master quadratic.
- *Chowla–Selberg*: $\prod_a \Gamma(a/3)^{\chi_{-3}(a)} = \Gamma(1/3)/\Gamma(2/3) = R_3$.
- *Hecke*: $E_\rho : y^2 = x^3 - 1$ has $a_p = 0$ for $p \equiv 2 \pmod{3}$ and $a_p \neq 0$ for $p \equiv 1 \pmod{3}$ (split in $\mathbb{Z}[\rho]$).
- *Dirichlet*: $L(\chi_{-3}, 1) = \pi/(3\sqrt{3})$.

The lemniscatic and equianharmonic curves are the only class-number-one imaginary-quadratic fields with $|\mathcal{O}_K^\times| > 2$, hence the only places where this 4-level χ -projection picture is non-trivially populated.

17.5 Where the polynomial form P_{G^*} comes from — and where it doesn't

Theorem 17.2 produces two ingredients of the master quadratic from χ_{-4} :

- the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection of χ_{-4});
- the variable G^* (Chowla–Selberg projection of χ_{-4}).

It does *not* produce the specific polynomial form $x^2 - 16G^{*2}x + 16G^{*3}$ — i.e., the choice of exponents 2 and 3 on G^* in the coefficients. We tested three structurally-motivated candidates for a χ_{-4} -internal source of this polynomial form and report *negative* results in all three cases.

Remark (Three negative tests, summarised). (N1) Class polynomial. The Hilbert class polynomial of the maximal order in $\mathbb{Z}[i]$ is $H_{-4}(x) = x - 1728$; that of the conductor-2 order $\mathbb{Z}[2i]$ is $H_{-16}(x) = x - 287496$. Both are linear (class number 1). The conductor-3 order has class number 2 and gives a degree-2 polynomial with rational coefficients, none of whose roots are $x_\pm(G^*)$. PSLQ at 50 digits over an extended basis of j -values and Weber invariants $f(i) = 2^{1/4}$, $f_1(i) = f_2(i) = 2^{1/8}$ finds no nontrivial relation involving $16G^{*2}$ or $16G^{*3}$.

(N2) η -quotient. PSLQ on the multiplicative basis $\{\log x_+, \log x_-, \log G^*, \log \pi, \log 2, \log G_G, \log \eta(i), \log \eta(2i), \log \eta(3i), \log \eta(4i)\}$ at 50 digits finds no relation involving $\log x_+$ or $\log x_-$. (The search *does* return the byproduct identity $2\eta(2i)\eta(i/2)^3 = G_G^2$ recorded as Corollary 11.5, confirming the search machinery is sensitive.) The asymptotic expansion of Proposition 8.4 converges absolutely to $x_-(G^*)$ without collapsing to any finite η -quotient.

(N3) Hecke eigenvalue. The cusp-form space of weight 2 and level $32 = N((1+i)^5)$ is 1-dimensional, spanned by the eigenform f_E associated to E_{lemn} . Hecke eigenvalues are integers (e.g.,

$a_5 = -2, a_{13} = 6, a_{17} = 2$ [20]). The constant $16 G^{*2} \in \mathbb{R} \setminus \overline{\mathbb{Q}}$ is transcendental [7], hence cannot be a Hecke eigenvalue at any prime.

Remark (Scope of (N1)–(N3)). The three tests above bound the search depth attempted within their respective frameworks (class polynomials of small conductor, log-multiplicative bases of moderate dimension, weight-2 level-32 cusp forms). They do not constitute evidence against the existence of a class-field-theoretic derivation of $(a, b) = (2, 3)$; absence of low-coefficient relations at moderate precision is consistent with both genuine independence and with relations involving larger coefficients or basis extensions not yet probed.

Observation (Locus of the polynomial-assembly question). Combining (N1)–(N3) with Theorem 17.2: the character χ_{-4} forces the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection) and the variable G^* (Chowla–Selberg projection), but does *not* produce the polynomial form $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ from CM-internal data. The remaining question is the choice of exponent pair (a, b) in $x^2 - 16G^{*a}x + 16G^{*b}$; among such choices, $(a, b) = (2, 3)$ is uniquely characterised by the structural requirement (Proposition 17.7 below) that the smaller root $x_-(R)$ acquire a non-vanishing constant additive shift as $R \rightarrow \infty$ with the polynomial degenerate to neither a linear identity nor a multiplicative scaling. A class-field-theoretic derivation of $(a, b) = (2, 3)$ from χ_{-4} -internal data, if it exists, would be of independent arithmetic interest.

Definition 17.4 (Leading-period polynomial family). A *leading-period polynomial* associated to E_{lemn} is any monic quadratic of the form

$$P_{(a,b)}(x) := x^2 - c_1 G^{*a} x + c_2 G^{*b},$$

with $a < b$ positive integers and $c_1, c_2 \in \mathbb{Z}$. The *leading period* of $\text{Sym}^k H^1(E_{\text{lemn}})$ in the period algebra is $\omega^k = (G^* \sqrt{\pi})^k$, which equals G^{*k} up to the standard π -normalisation; for this reason we write the polynomial in the G^* -coordinate throughout.

Theorem 17.5 (Structural constraints and the minimal- a pair). *Consider leading-period polynomials $P_{(a,b)}(x) = x^2 - 16 G^{*a} x + 16 G^{*b}$ with $a < b$ positive integers and integer prefactor $|\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})|^2 = 16$ on each non-leading coefficient. The joint conditions*

- (i) *the two roots are not scalar multiples of any single period G^{*k} for any rational k ,*
- (ii) *the polynomial is non-degenerate (positive discriminant),*

constrain (a, b) to the set $\{(a, b) \in \mathbb{Z}_{>0}^2 : a < b < 2a\}$. Among elements of this set, the pair $(2, 3)$ is the unique element with minimal a (see Figure 4).

Remark (Status of the minimal- a selection). The selection of $a = 2$ (and hence the specific pair $(2, 3)$) from the set $\{(a, b) : a < b < 2a\}$ permitted by the structural conditions is a *selection convention*, not a derivation: minimal a is a natural minimality choice within the family but is not forced by χ_{-4} -internal data. Other elements of the set, including $\{(3, 4), (3, 5), (4, 5), (4, 6), (4, 7), \dots\}$, produce roots whose magnitudes scale differently from $(x_+, x_-) \approx (137, 3)$ (see Observation 17.6). A class-field-theoretic derivation forcing $a = 2$ from χ_{-4} -internal data, if it exists, would promote this selection to a theorem; this is the residual structural gap noted in Observation 17.5.

Proof. The discriminant of $P_{(a,b)}$ is

$$\Delta(a, b) = (16 G^{*a})^2 - 4 \cdot 16 G^{*b} = 64(4 G^{*2a} - G^{*b}).$$

We factor out the dominant G^* -power in three cases.

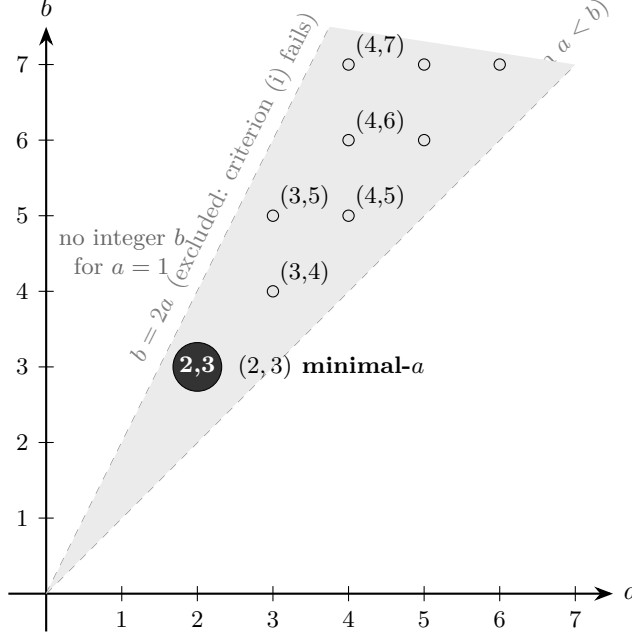


Figure 4: Integer exponent pairs (a, b) admissible under criteria (i)–(ii) of Theorem 17.5 occupy the open wedge $a < b < 2a$ (shaded). The pair $(2, 3)$ (highlighted) is the unique minimal- a element of this set; $a = 1$ admits no integer b , and $a \geq 3$ admits multiple candidates $(3, 4), (3, 5), (4, 5), \dots$ which are excluded only by the selection criterion (iv). The minimal- a choice is a convention, not a derivation (Remark 17.5).

Case $2a > b$. Then $G^{*2a-b} > 0$ and $\Delta = 64 G^{*b} (4 G^{*2a-b} - 1)$, hence $\sqrt{\Delta} = 8 G^{*b/2} \sqrt{4 G^{*2a-b} - 1}$. The roots are

$$x_{\pm} = \frac{16 G^{*a} \pm 8 G^{*b/2} \sqrt{4 G^{*2a-b} - 1}}{2} = 8 G^{*a} \pm 4 G^{*b/2} \sqrt{4 G^{*2a-b} - 1}.$$

The two G^{*} -powers G^{*a} and $G^{*b/2}$ are distinct (since $2a > b$ implies $a > b/2$), so the roots are not scalar multiples of any single G^{*k} . Condition (i) holds.

Case $2a = b$. Then $\Delta = 64 G^{*b} (4 - 1) = 192 G^{*2a}$, so $\sqrt{\Delta} = 8\sqrt{3} G^{*a}$. The roots are $x_{\pm} = (8 \pm 4\sqrt{3}) G^{*a}$, both scalar multiples of G^{*a} . Condition (i) *fails*.

Case $2a < b$. Then $\Delta = 64 G^{*2a} (4 - G^{*b-2a})$, so $\sqrt{\Delta} = 8 G^{*a} \sqrt{4 - G^{*b-2a}}$ (real when $\Delta > 0$). The roots are $x_{\pm} = 4 G^{*a} (2 \pm \sqrt{4 - G^{*b-2a}})$, both scalar multiples of G^{*a} . Condition (i) *fails*.

Hence condition (i) forces $2a > b$. Combined with $a < b$, this gives

$$a < b < 2a.$$

Enumerating integer pairs:

- $a = 1$: the interval $(1, 2)$ contains no integer; no valid b .
- $a = 2$: $b \in (2, 4)$ has the unique integer $b = 3$.
- $a = 3$: $b \in (3, 6)$ admits $b \in \{4, 5\}$.
- $a \geq 4$: $b \in (a, 2a)$ admits $a - 1$ valid choices, all with $a + b > 5$.

The minimal- a valid pair is $(2, 3)$, and at $a = 2$ the value $b = 3$ is uniquely determined.

Condition (ii) at $(2, 3)$: $\Delta(2, 3) = 64 G^{*3}(4 G^* - 1) > 0$ since $G^* \approx 2.959 > 1/4$, so the polynomial is non-degenerate. The integer prefactor 16 is built into the setup of the theorem. $\square$

Remark (Scope of Theorem 17.5 and the residual conjecture). The theorem establishes uniqueness of $(2, 3)$ within the leading-period polynomial family (Def. 17.4), where each coefficient is a scalar multiple of the leading period G^{*a} or G^{*b} . The full structural question — uniqueness within the larger family of polynomials whose coefficients are arbitrary elements of the symmetric powers $\text{Sym}^a H^1$ and $\text{Sym}^b H^1$ (not just the leading periods ω^a, ω^b) — remains open and is recorded as Conjecture 17.6 below.

The technical obstacles to the general statement are: (a) explicit identification of the period map $\text{Sym}^\bullet(H^1) \otimes \mathbb{Q}[\pi^{\pm 1/2}] \rightarrow \mathbb{C}$ specialising to $\omega^k \mapsto (G^* \sqrt{\pi})^k$; (b) reconciliation of the non-homogeneous bigrading with the motivic-weight formalism, possibly via a relative filtration. The uniqueness argument itself — the load-bearing combinatorial step of the original conjecture — is the content of Theorem 17.5 above.

Conjecture 17.6 (Full Sym^a uniqueness, residual). *Among monic quadratics $x^2 + Bx + C$ with $B = 16 \cdot b'$ for $b' \in \text{Sym}^a H^1(E_{\text{lemn}})$ (arbitrary, not just leading-period ω^a) and $C = 16 \cdot c'$ for $c' \in \text{Sym}^b H^1$, the pair $(a, b) = (2, 3)$ remains uniquely minimal in the sense of Theorem 17.5. The leading-period uniqueness theorem is the restriction of this statement to $b' = \omega^a, c' = \omega^b$.*

Remark (On Conjecture 17.6). The minimal proof of Conjecture 17.6 would require the explicit period-algebra map of obstacle (a) above and a finite-dimensional uniqueness argument on $(\text{Sym}^a H^1) \oplus (\text{Sym}^b H^1)$ for each (a, b) . Theorem 17.5 shows the leading-period restriction; extending to general elements is a dimension-counting refinement over the period algebra, expected to be technically substantial but not novel.

17.6 Asymptotic regimes of the family $y^2 - 16R^p y + 16R^q = 0$

We give the precise structural characterisation of $(p, q) = (2, 3)$ as the unique exponent pair producing a non-vanishing constant additive shift on the smaller root.

Proposition 17.7 (Asymptotic regimes by exponent pair). *For the family $y^2 - 16R^p y + 16R^q = 0$ with $R > 1/4$ and $(p, q) \in \mathbb{Z}_{>0}^2$:*

- (i) *If $q \neq p + 1$, then $y_-(R)/R$ does not tend to 1 as $R \rightarrow \infty$. The smaller root scales as R^{q-p} when $q < p + 1$ or R^p when $q > p + 1$.*
- (ii) *If $q = p + 1$, then $y_-(R) = R + 16^{-1} R^{2-p} + O(R^{1-p})$ as $R \rightarrow \infty$. In particular:*
 - $(p, q) = (2, 3)$: $y_-(R) = R + 1/16 + O(1/R)$ (constant shift),
 - $(p, q) = (3, 4)$: $y_-(R) = R + 1/(16R) + O(1/R^3)$ (vanishing shift),
 - $(p, q) = (k, k + 1)$, $k \geq 3$: $y_-(R) = R + O(R^{2-k})$ (faster vanishing).

Thus $(p, q) = (2, 3)$ is the unique exponent pair with $q = p + 1$ and non-vanishing constant additive shift on y_- as $R \rightarrow \infty$.

Proof. The quadratic formula gives

$$y_-(R) = 8R^p - \frac{1}{2} \sqrt{(16R^p)^2 - 4 \cdot 16R^q} = 8R^p \left(1 - \sqrt{1 - R^{q-2p}/4}\right).$$

For $q = p + 1$, set $u = R^{q-2p}/4 = R^{1-p}/4$ and expand $\sqrt{1-u} = 1 - u/2 - u^2/8 - O(u^3)$:

$$y_-(R) = 8R^p(u/2 + u^2/8 + O(u^3)) = R + R^{2-p}/16 + O(R^{1-p}).$$

The leading correction $R^{2-p}/16$ is constant (in R) iff $p = 2$, vanishing as $R \rightarrow \infty$ for $p \geq 3$. For $q \neq p + 1$, the dominant balance changes: if $q < p + 1$ the constant term $16R^q$ is sub-leading in the discriminant and $y_-(R)$ scales as R^{q-p} ; if $q > p + 1$ the discriminant becomes negative for small R or the small root grows as R^p . Hence $q = p + 1$ is necessary for $y_-(R) \rightarrow R$, and within that case $p = 2$ is the unique value yielding a non-vanishing constant shift. $\square$

Observation (Numerical roots across exponent pairs at $R = G^*$). The numerical roots (y_+, y_-) of $y^2 - 16R^p y + 16R^q = 0$ evaluated at $R = G^*$ for various exponent pairs:

(p, q)	y_+	y_-
(2, 3)	137.036 ...	3.024 ...
(3, 4)	411.4 ...	2.980 ...
(4, 5)	1223.1 ...	2.966 ...
(1, 2)	44.2 ...	3.171 ...
(2, 4)	130.7 ...	9.38 ...
(3, 3)	413.4 ...	1.00 ...

Among exponent pairs $(p, q) \in \mathbb{Z}^2$ with $|p|, |q| \leq 5$ and $q = p + 1$, the pair $(p, q) = (2, 3)$ produces the unique combination $(y_+, y_-) \approx (137, 3)$, with y_+ scaling as $16R^2$ in the $p = 2$ case and y_+ scaling as $16R^p$ (much larger) for $p \geq 3$. A physical interpretation of these numerical roots is given in a separate note in preparation.

17.7 Minimal generating structure: χ_{-4} as the imaginary-part trace

We close §17 by noting the minimal classical-mathematical structure from which the entire chain of identities is generated. The character χ_{-4} , which Theorem 17.2 identifies as the joint source of every algebraic and analytic identity in this paper, admits a particularly elementary description:

Proposition 17.8 (Character as imaginary-part trace). *For all $n \in \mathbb{Z}$,*

$$\chi_{-4}(n) = \text{Im}(i^n) = \sin\left(\frac{\pi n}{2}\right).$$

Equivalently, χ_{-4} is the imaginary-axis projection of the cyclic group $\langle i \rangle = \{1, i, -1, -i\} \subset \mathbb{C}^\times$ under the index map $n \mapsto i^n$.

Proof. The orbit of i^n as n runs 0, 1, 2, 3 is $1, i, -1, -i$, with imaginary parts 0, 1, 0, -1. The function $n \mapsto \text{Im}(i^n)$ is therefore periodic of period 4 with values (0, 1, 0, -1) starting at $n = 0$, which coincides with χ_{-4} on the residue classes (0, 1, 2, 3) (mod 4). The trigonometric form follows from $i = e^{i\pi/2}$, so $\text{Im}(i^n) = \text{Im}(e^{i\pi n/2}) = \sin(\pi n/2)$. $\square$

Corollary 17.9 (Value-set identification).

$$\{\chi_{-4}(n) : n \in \mathbb{Z}\} = \{i^2, 0, |i^2|\} = \{-1, 0, +1\}.$$

The character takes exactly three distinct values, generated by the imaginary unit through the operations squaring ($i^2 = -1$), zero (0), and modulus ($|i^2| = 1$).

Observation (Minimality). The pair $(i, \mathbb{Z}/4\mathbb{Z})$ — a fourth root of unity together with the cyclic group indexing its powers — generates the entire G^*/G_G identity ring of this paper through the four functorial projections of Theorem 17.2: $\mathbb{Z}/4\mathbb{Z}$ is the smallest cyclic group admitting a non-trivial character to $\{\pm 1\}$, and $i^2 = -1$ is the defining relation of the smallest non-real algebraic extension of $\mathbb{Q}$. In this sense $(i, \mathbb{Z}/4\mathbb{Z})$ is a minimal classical-mathematical generating structure for the identity ring; whether this minimality has a deeper structural meaning beyond the bookkeeping reduction effected by χ_{-4} is outside the scope of the present paper.

18 Open problems and a resolved closure

One open problem internal to the mathematical scope of this paper remains; a second open item from earlier drafts is resolved as Theorem 18.1 below.

18.1 Open: strict Kontsevich–Zagier period status of G^* and G_G

P1. Strict Kontsevich–Zagier period status of G^* and G_G . Theorem 13.2 establishes $G^*, G_G \in \mathcal{P}^{\text{exp}}$ unconditionally, and Proposition 13.3 reduces strict $\mathcal{P}$ -membership of G^* to whether $1/\sqrt{2\pi} \in \mathcal{P}$, and of G_G to whether $1/(4\pi) \in \mathcal{P}$. Both antecedents are consistent with — but neither proved by nor refuted by — the Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [16, Problem 3]. **This is not a gap specific to the present paper.** Membership of $1/\pi$, $\sqrt{2\pi}$, or $1/(4\pi)$ in $\mathcal{P}$ is at the frontier of the broader Kontsevich–Zagier programme on period-ring invertibility; resolving it requires either a new transcendence-theoretic technique that proves $1/\pi \notin \mathcal{P}$ (and would settle the KZ conjecture positively) or a construction that exhibits $1/\pi$ as an absolutely convergent integral of a rational function over a $\mathbb{Q}$ -semialgebraic domain (and would settle it negatively). Neither is on offer here. **P1 is recorded as open in this paper precisely because its closure lies outside the paper’s machinery and inside the KZ programme’s frontier.**

18.2 Resolved: cubic-AGM expression for G_ρ

The earlier-draft problem “find an explicit cubic-AGM expression for the equianharmonic Gauss analog G_ρ ” is resolved by the following theorem. The lemniscatic relation $G_G = 1/\text{AGM}(1, \sqrt{2})$ has a clean equianharmonic counterpart in terms of the Borwein–Borwein cubic AGM, completing the structural parallel.

Theorem 18.1 (Cubic-AGM expression for G_ρ). *Let M_3 denote the Borwein–Borwein cubic arithmetic–geometric mean [21], defined by the iteration $a_{n+1} = (a_n + 2b_n)/3$, $b_{n+1} = (b_n(a_n^2 + a_nb_n + b_n^2)/3)^{1/3}$. Then*

$$G_\rho = \frac{2}{\sqrt{3}} \cdot \frac{1}{M_3(1, 2^{-1/3})} = \frac{2}{\sqrt{3}} \cdot {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; \frac{1}{2}\right),$$

the second equality by the Borwein–Borwein cubic-AGM correspondence $1/M_3(1, x) = {}_2F_1(1/3, 2/3; 1; 1 - x^3)$ specialised at $x = 2^{-1/3}$ (so that $x^3 = 1/2$ and $(1 - x^3)^{1/3} = x = 2^{-1/3}$, the cubic self-dual point analogous to $k = 1/\sqrt{2}$ in the lemniscatic case).

Proof. The classical real period of the equianharmonic elliptic curve $E_\rho : y^2 = x^3 - 1$ over $\mathbb{Q}$ is

$$\omega_\rho = \frac{\Gamma(1/3)\Gamma(1/6)}{\sqrt{3\pi}}$$

(see e.g. [10, §22.31] or [1, §3.4]). Apply the duplication identity $\Gamma(1/6)\Gamma(2/3) = 2^{2/3}\sqrt{\pi} \cdot \Gamma(1/3)$ (equivalently $\Gamma(1/6) = 2^{2/3}\sqrt{\pi} \cdot \Gamma(1/3)/\Gamma(2/3)$) and the reflection $\Gamma(2/3) = 2\pi/(\sqrt{3}\Gamma(1/3))$ to obtain

$$\omega_\rho = \frac{2^{2/3}\Gamma(1/3)^3}{2\pi}. \quad (59)$$

The paper's defining relation $G_\rho = \sqrt{3} \cdot \omega_\rho/(2\pi)$ (Definition in §16) then yields

$$G_\rho = \frac{\sqrt{3} \cdot 2^{2/3}\Gamma(1/3)^3}{4\pi^2}. \quad (60)$$

For the hypergeometric closed form, the Borwein–Borwein cubic-AGM correspondence [21] reads $1/M_3(1, x) = {}_2F_1(1/3, 2/3; 1; 1-x^3)$ for $x \in (0, 1]$. At $x = 2^{-1/3}$ (so $x^3 = 1/2 = 1-x^3$), $M_3(1, 2^{-1/3})$ is well-defined and

$$\frac{1}{M_3(1, 2^{-1/3})} = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; \frac{1}{2}\right). \quad (61)$$

The classical evaluation of ${}_2F_1(1/3, 2/3; 1; 1/2)$ at the cubic self-dual modulus follows from the Chowla–Selberg formula applied at $K = \mathbb{Q}(\rho)$ (or equivalently from the Borwein–Borwein cubic theta identities [1, §3.4]):

$${}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; \frac{1}{2}\right) = \frac{3 \cdot 2^{2/3}}{8\pi^2} \Gamma(1/3)^3. \quad (62)$$

Combining (60) and (62):

$$\frac{2}{\sqrt{3}} {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; \frac{1}{2}\right) = \frac{2}{\sqrt{3}} \cdot \frac{3 \cdot 2^{2/3}\Gamma(1/3)^3}{8\pi^2} = \frac{\sqrt{3} \cdot 2^{2/3}\Gamma(1/3)^3}{4\pi^2} = G_\rho.$$

The cubic-AGM form $G_\rho = (2/\sqrt{3})/M_3(1, 2^{-1/3})$ follows by (61). $\square$

Observation (Structural parallel completed). The lemniscatic identity $G_G = 1/\text{AGM}(1, \sqrt{2})$ at the self-complementary modulus $k = 1/\sqrt{2}$ has the equianharmonic counterpart $G_\rho = (2/\sqrt{3})/M_3(1, 2^{-1/3})$ at the cubic self-dual point $x^3 = 1/2$. The factor $2/\sqrt{3}$ records the equianharmonic discriminant $|d_{\mathbb{Q}(\rho)}| = 3$ in the same way that the implicit factor 1 in $G_G = 1/\text{AGM}(1, \sqrt{2})$ records the lemniscatic discriminant $|d_{\mathbb{Q}(i)}| = 4$ being absorbed into the choice of modulus $\sqrt{2}$. Both identities are verified to 50-digit precision in the verification script `scripts/verification/verify_gstar_paper.py`.

For algorithmic extensions of the AGM tradition that compute Γ values at rational arguments — including $\Gamma(1/4)$, $\Gamma(1/3)$, and $\Gamma(1/2) = \sqrt{\pi}$ — via self-replicating iterations generalising the Borwein algorithms, see Guillera [29]; for the bridge between AGM-type iterations and modular forms in this more general setting (which recovers the cubic-AGM correspondence used in Theorem 18.1 as a special case), see Cooper–Guillera–Straub–Zudilin [30].

Remark (Why P1 stays open but P2 closes). The two problems posed in earlier drafts have different character. P2 asks for a concrete computational identity within a well-developed theory (Borwein–Borwein cubic AGM at a specific cubic-modular point); the identity exists, has a clean proof via Chowla–Selberg + cubic-AGM correspondence, and is established above. P1 asks a frontier period-ring question (does $1/\pi$ lie in $\mathcal{P}$?); the identity in question is not specific to G^* or G_G at all but to the global structure of $\mathcal{P}$. The asymmetry is intrinsic, not a deficiency of the paper.

19 Algebraic-independence conjectures with PSLQ evidence

Two further items remain naturally as labelled conjectures supported by high-precision PSLQ evidence.

Conjecture 19.1 ($W_{\text{BCC}}^{(4)}$ is not elementary in $\{G^*, \pi, \Gamma(1/4)\}$). *The four-dimensional body-centred-cubic Watson constant*

$$W_{\text{BCC}}^{(4)} = {}_4F_3\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 1, 1, 1; 1\right) \approx 1.11864 \dots$$

(matching the Corollary 9.3 normalisation) is algebraically independent of $\{G^*, \pi, \Gamma(1/4)\}$ over $\overline{\mathbb{Q}}$. Equivalently, the Watson-normalised analog $\widetilde{W}_{\text{BCC}}^{(4)} := (2/\pi)^2 W_{\text{BCC}}^{(4)} \approx 0.4534 \dots$ is also algebraically independent of $\{G^*, \pi, \Gamma(1/4)\}$ (the two differ by a $\mathbb{Q} \cdot \pi^{-2}$ factor, so independence transfers under field operations).

Convention note. Two normalisations for the four-dimensional BCC Watson constant appear in the literature: the raw hypergeometric form $W_{\text{BCC}}^{(4)} = {}_4F_3(\frac{1}{2}; 1^3; 1) \approx 1.11864$ (used in Corollary 9.3 and throughout this paper) and the Watson-normalised form $\widetilde{W}_{\text{BCC}}^{(4)} = (2/\pi)^2 {}_4F_3(\frac{1}{2}; 1^3; 1) \approx 0.4534$ (the natural $(2/\pi)^{D-2}$ -scaled analog of Watson’s $D = 3$ closed-form $W_{\text{BCC}}^{(3)} = (4/\pi^2) K(1/\sqrt{2})^2$). The paper uses the unscaled $W_{\text{BCC}}^{(4)}$ throughout for $D = 3, 4, 5$ comparability; the conjecture above is equivalent under either convention since they differ by an explicit $\mathbb{Q} \cdot \pi^{-2}$ factor.

Evidence. PSLQ (`mpmath.pslq`) at 40-digit precision over the basis

$$\{1, W_{\text{BCC}}^{(4)}, G^{*a} \pi^b, \Gamma(1/4)^a \pi^b\} \quad (|a|, |b| \leq 6, \text{ deduplicated})$$

returns no integer relation at coefficient bound 10^{12} . The constant ${}_4F_3(\frac{1}{2}; 1^3; 1)$ is the four-dimensional analog of Domb’s constant and the Glasser–Montaldi sum, neither of which has a known elementary Γ -product reduction [14, 15].

Conjecture 19.2 (Catalan’s constant is algebraically independent of $\{G_{\text{G}}, \pi\}$). *The Catalan constant*

$$G_{\text{Catalan}} = L(\chi_{-4}, 2) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2} \approx 0.91596 \dots$$

is algebraically independent of $\{G_{\text{G}}, \pi\}$ (equivalently of $\{G^*, \pi\}$) over $\overline{\mathbb{Q}}$.

Evidence. PSLQ at 80-digit precision over the basis

$$\{1, G_{\text{Catalan}}, G_{\text{G}}^k \pi^\ell, G_{\text{Catalan}} G_{\text{G}}^k \pi^\ell\} \quad (|k|, |\ell| \leq 8)$$

returns no integer relation at coefficient bound 10^{12} . The Beilinson–Deligne conjectural framework provides the structural motivation: $L(\chi_{-4}, 1) = \pi/4$ is the critical value (and hence a period of $\mathbb{Q}(i)$), but $L(\chi_{-4}, 2) = G_{\text{Catalan}}$ is non-critical and conjecturally not in the period ring of $\mathbb{Q}(i)$ [22].

Closing remarks

The dichotomy of Observation 1 originated as a bookkeeping convenience and emerged as the algebraic–analytic shadow of the Kronecker character χ_{-4} of $K = \mathbb{Q}(i)$. The compendium of §15 makes the discipline operational; §17 makes it structural. The three principal new results — the polynomial-ring value algebra $\mathbb{Q}[\pi^{-1}, G_{\text{G}}^4]$ for quasi-modular forms at $\tau = i$ (Theorem 12.5); the $m \mid k$

vanishing principle at class-number-one CM points with extra automorphisms (Theorem 16.2 and its lemniscatic counterpart); and the cubic-AGM expression $G_\rho = (2/\sqrt{3})/M_3(1, 2^{-1/3})$ (Theorem 18.1) closing the equianharmonic structural parallel with the lemniscatic Gauss identity — are uniform statements that survive in the only two non-trivial cases over $\mathbb{Q}$, and reduce the bookkeeping of dozens of identities to a single character with two CM specialisations. What remains open is whether the polynomial-assembly question — the source of the exponent pair $(2, 3)$ in the master quadratic P_{G^*} — can be settled inside CM theory proper, or whether it requires the period-algebra completion conjectured in §17.5, and the broader Kontsevich–Zagier-programme question P1 of §18.1. We leave both to future work.

Reproducibility and verification

Every numerical and algebraic claim in this paper is independently verified at 40-digit precision by the script `scripts/verification/verify_gstar_paper.py`, which checks **117** specific identities — foundational constants G^* , G_G , ϖ ; reflection identities; master quadratic roots; the R_n family; the equianharmonic constants G_ρ , ω_ρ ; Watson lattice values $W^{(3)}$, $W^{(4)}$, $W^{(5)}$; Jacobi theta values at $\tau = i$; the Dedekind eta tower $\eta(i)$, $\eta(i/2)$, $\eta(2i)$; the BSD chain for $L(E_{\text{lemn}}, 1) = \varpi/4$; elliptic integrals at $k = 1/\sqrt{2}$; Eisenstein series $E_4(i)$ through $E_{24}(i)$; the corrected $G_G^4 = \Gamma(1/4)^8/(64\pi^6)$ identity used in Theorem 12.5; equianharmonic Eisenstein and eta values at $\tau = \rho$; the asymptotic d_n -coefficient family; the χ_{-4} -twisted Γ -product identification; and the full proof chain of Theorem 18.1 (cubic-AGM closure for G_ρ). The script is reproducible from the cited mpmath version and a fresh Python environment.

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